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Bed interface for manual location

Abstract

A graphical user interface (GUI) of a patient bed is used by a caregiver to manually enter location data indicative of a location in a healthcare facility at which the patient bed is located. A first screen for manually entering location data is displayed on the GUI after a threshold period of time elapses subsequent to a power plug of the patient bed being plugged into an alternating current (AC) outlet of the healthcare facility and subsequent to casters of the patient bed being braked. A voice prompt is also given from the patient bed after the threshold period of time elapses to remind the caregiver to manually enter the location data. After manual entry of the location data, circuitry of the patient bed transmits the location data entered by the caregiver and a bed identification (ID) from the bed for receipt by a remote computer for purposes of making a bed-to-room association.

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6294953	12/2000	Steeves	N/A	N/A
6302844	12/2000	Walker et al.	N/A	N/A
6314556	12/2000	DeBusk et al.	N/A	N/A
6320510	12/2000	Menkedick et al.	N/A	N/A
RE37467	12/2000	Brasch et al.	N/A	N/A
6333690	12/2000	Nelson et al.	N/A	N/A
RE37531	12/2001	Chaco et al.	N/A	N/A
6343064	12/2001	Jabbarnezhad	N/A	N/A
6344794	12/2001	Ulrich et al.	N/A	N/A
6348777	12/2001	Brown et al.	N/A	N/A
6353413	12/2001	White et al.	N/A	N/A
6362725	12/2001	Ulrich et al.	N/A	N/A
6364834	12/2001	Reuss et al.	N/A	N/A
6398727	12/2001	Bui et al.	N/A	N/A
6400956	12/2001	Richton	N/A	N/A
6406426	12/2001	Reuss et al.	N/A	N/A
6407335	12/2001	Franklin-Lees et al.	N/A	N/A
6412980	12/2001	Lounsberry et al.	N/A	N/A
6416471	12/2001	Kumar et al.	N/A	N/A
6418416	12/2001	Rosenberg et al.	N/A	N/A
6421649	12/2001	Rattner	N/A	N/A
6424264	12/2001	Giraldin et al.	N/A	N/A
6439769	12/2001	Polkus et al.	N/A	N/A
6441742	12/2001	Lovely et al.	N/A	N/A
6442290	12/2001	Ellis et al.	N/A	N/A
6445299	12/2001	Rojas, Jr.	N/A	N/A
6450956	12/2001	Rappaport et al.	N/A	N/A
6462656	12/2001	Ulrich et al.	N/A	N/A
6486792	12/2001	Moster et al.	N/A	N/A
6486794	12/2001	Calistro et al.	N/A	N/A
6493568	12/2001	Bell et al.	N/A	N/A
6494831	12/2001	Koritzinsky	N/A	N/A
6510344	12/2002	Halpern	N/A	N/A
6516324	12/2002	Jones et al.	N/A	N/A
6526310	12/2002	Carter et al.	N/A	N/A
6529164	12/2002	Carter	N/A	N/A
6533453	12/2002	Heidsieck et al.	N/A	N/A
6535576	12/2002	Vafi et al.	N/A	N/A
6544174	12/2002	West et al.	N/A	N/A
6551243	12/2002	Bocionek et al.	N/A	N/A
6553106	12/2002	Gould et al.	N/A	N/A
6554174	12/2002	Aceves	N/A	N/A

6556630	12/2002	Brinsfield et al.	N/A	N/A
6560274	12/2002	Leitgeb et al.	N/A	N/A
6572556	12/2002	Stoycos et al.	N/A	N/A
6575901	12/2002	Stoycos et al.	N/A	N/A
6581204	12/2002	DeBusk et al.	N/A	N/A
6584182	12/2002	Brodnick	N/A	N/A
6584454	12/2002	Hummel, Jr. et al.	N/A	N/A
6585645	12/2002	Hutchinson	N/A	N/A
6589170	12/2002	Flach et al.	N/A	N/A
6593528	12/2002	Franklin-Lees et al.	N/A	N/A
6593845	12/2002	Friedman et al.	N/A	N/A
6594146	12/2002	Frangesch et al.	N/A	N/A
6594519	12/2002	Stoycos et al.	N/A	N/A
6600418	12/2002	Francis et al.	N/A	N/A
6600421	12/2002	Freeman	N/A	N/A
6603494	12/2002	Banks et al.	N/A	N/A
6609115	12/2002	Mehring et al.	N/A	N/A
6616606	12/2002	Petersen et al.	N/A	N/A
6622088	12/2002	Hood	N/A	N/A
6640246	12/2002	Gary, Jr. et al.	N/A	N/A
6643238	12/2002	Nakajima	N/A	N/A
6650346	12/2002	Jaeger et al.	N/A	N/A
6659947	12/2002	Carter et al.	N/A	N/A
6665385	12/2002	Rogers et al.	N/A	N/A
6665820	12/2002	Frowein et al.	N/A	N/A
6669630	12/2002	Joliat et al.	N/A	N/A
6671547	12/2002	Lyster et al.	N/A	N/A
6671563	12/2002	Engelson et al.	N/A	N/A
6685633	12/2003	Albert et al.	N/A	N/A
6689091	12/2003	Bui et al.	N/A	N/A
6693513	12/2003	Tuttle	N/A	N/A
6693514	12/2003	Perea, Jr. et al.	N/A	N/A
6694367	12/2003	Miesbauer et al.	N/A	N/A
6694509	12/2003	Stoval et al.	N/A	N/A
6697765	12/2003	Kuth	N/A	N/A
6707476	12/2003	Hochstedler	N/A	N/A
6714913	12/2003	Brandt et al.	N/A	N/A
6721818	12/2003	Nakamura	N/A	N/A
6726634	12/2003	Freeman	N/A	N/A
6727818	12/2003	Wildman et al.	N/A	N/A
6731311	12/2003	Bufe et al.	N/A	N/A
6731908	12/2003	Berliner et al.	N/A	N/A
6731989	12/2003	Engleson et al.	N/A	N/A
6740033	12/2003	Olejniczak et al.	N/A	N/A
6747560	12/2003	Stevens, III	N/A	N/A
6747562	12/2003	Giraldin et al.	N/A	N/A
6749566	12/2003	Russ	N/A	N/A
6751630	12/2003	Franks et al.	N/A	N/A
6754545	12/2003	Haeuser et al.	N/A	N/A
6754883	12/2003	DeBusk et al.	N/A	N/A

6756918	12/2003	Fomukong	N/A	N/A
6759959	12/2003	Wildman	N/A	N/A
6763541	12/2003	Mahoney et al.	N/A	N/A
6771172	12/2003	Robinson et al.	N/A	N/A
6773396	12/2003	Flach et al.	N/A	N/A
6778225	12/2003	David	N/A	N/A
6781517	12/2003	Moster et al.	N/A	N/A
6784797	12/2003	Smith et al.	N/A	N/A
6791460	12/2003	Dixon et al.	N/A	N/A
6792396	12/2003	Inda et al.	N/A	N/A
6798533	12/2003	Tipirneni	N/A	N/A
6801227	12/2003	Bocionek et al.	N/A	N/A
6807543	12/2003	Muthya	N/A	N/A
6812824	12/2003	Goldinger et al.	N/A	N/A
6825763	12/2003	Ulrich et al.	N/A	N/A
6826578	12/2003	Brackett et al.	N/A	N/A
6828992	12/2003	Freeman et al.	N/A	N/A
6829796	12/2003	Salvatini et al.	N/A	N/A
6830549	12/2003	Bui et al.	N/A	N/A
6832199	12/2003	Kucek et al.	N/A	N/A
6838992	12/2004	Tenarvitz	N/A	N/A
6840117	12/2004	Hubbard, Jr.	N/A	N/A
6843415	12/2004	Vogler	N/A	N/A
6847435	12/2004	Honda et al.	N/A	N/A
6847814	12/2004	Vogeleisen	N/A	N/A
6859761	12/2004	Bensky et al.	N/A	N/A
6868256	12/2004	Dooley et al.	N/A	N/A
6871211	12/2004	Labounty et al.	N/A	N/A
6873884	12/2004	Brackett et al.	N/A	N/A
6876303	12/2004	Reeder et al.	N/A	N/A
6876985	12/2004	Kawanaka	N/A	N/A
6884255	12/2004	Newton	N/A	N/A
6885288	12/2004	Pincus	N/A	N/A
6891909	12/2004	Hurley et al.	N/A	N/A
6897780	12/2004	Ulrich et al.	N/A	N/A
6904161	12/2004	Becker et al.	N/A	N/A
6909995	12/2004	Shiraishi	N/A	N/A
6912549	12/2004	Rotter et al.	N/A	N/A
6915170	12/2004	Engleson et al.	N/A	N/A
6925367	12/2004	Fontius	N/A	N/A
6930878	12/2004	Brackett et al.	N/A	N/A
6954148	12/2004	Pulkkinen et al.	N/A	N/A
6958706	12/2004	Chaco et al.	N/A	N/A
6968194	12/2004	Aljadeff et al.	N/A	N/A
6972682	12/2004	Lareau et al.	N/A	N/A
6988989	12/2005	Weiner et al.	N/A	N/A
6998985	12/2005	Reisman et al.	N/A	N/A
6998986	12/2005	Smith	N/A	N/A
7002468	12/2005	Eveland et al.	N/A	N/A
7003443	12/2005	Ford et al.	N/A	N/A

7005980	12/2005	Schmidt et al.	N/A	N/A
7009495	12/2005	Hughes et al.	N/A	N/A
7009516	12/2005	Enea	N/A	N/A
7014100	12/2005	Zierolf	N/A	N/A
7019663	12/2005	Sharony	N/A	N/A
7030761	12/2005	Bridgelall et al.	N/A	N/A
7030811	12/2005	Goren et al.	N/A	N/A
7034684	12/2005	Boman et al.	N/A	N/A
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7035818	12/2005	Bandy et al.	N/A	N/A
7038573	12/2005	Bann	N/A	N/A
7038584	12/2005	Carter	N/A	N/A
7038588	12/2005	Boone et al.	N/A	N/A
7038589	12/2005	Schmidt et al.	N/A	N/A
7042358	12/2005	Moore	N/A	N/A
7042361	12/2005	Kazdin et al.	N/A	N/A
7044387	12/2005	Becker et al.	N/A	N/A
7046153	12/2005	Oja et al.	N/A	N/A
7046162	12/2005	Dunstan	N/A	N/A
7049594	12/2005	Wu et al.	N/A	N/A
7053779	12/2005	Thompson	N/A	N/A
7053781	12/2005	Haire et al.	N/A	N/A
7053831	12/2005	Dempsey et al.	N/A	N/A
7056289	12/2005	Kasper et al.	N/A	N/A
7057509	12/2005	Gualdi et al.	N/A	N/A
7061366	12/2005	Bell et al.	N/A	N/A
7061384	12/2005	Fujimoto	N/A	N/A
7062455	12/2005	Tobey	N/A	N/A
7068143	12/2005	Doering et al.	N/A	N/A
7071820	12/2005	Callaway	N/A	N/A
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7075438	12/2005	Kent et al.	N/A	N/A
7080061	12/2005	Kabala	N/A	N/A
7084740	12/2005	Bridgelall	N/A	N/A
7091879	12/2005	Swetlik et al.	N/A	N/A
7092376	12/2005	Schuman	N/A	N/A
7099895	12/2005	Dempsey	N/A	N/A
7102510	12/2005	Boling et al.	N/A	N/A
7106189	12/2005	Burneske et al.	N/A	N/A
7116230	12/2005	Klowak	N/A	N/A
7117027	12/2005	Zheng et al.	N/A	N/A
7138913	12/2005	Mackenzie et al.	N/A	N/A
7142112	12/2005	Buckingham et al.	N/A	N/A
7151455	12/2005	Lindsay et al.	N/A	N/A
7152791	12/2005	Chappidi et al.	N/A	N/A
7154397	12/2005	Zerhusen et al.	N/A	N/A
7158030	12/2006	Chung	N/A	N/A
7164354	12/2006	Panzer	N/A	N/A
7165040	12/2006	Ehrman et al.	N/A	N/A
7167095	12/2006	Carrender	N/A	N/A

7170407	12/2006	Wagner	N/A	N/A
7174172	12/2006	Sharony et al.	N/A	N/A
7190778	12/2006	Kucmerowski	N/A	N/A
7196621	12/2006	Kochis	N/A	N/A
7199716	12/2006	Shanks et al.	N/A	N/A
7202785	12/2006	Maloney	N/A	N/A
7203690	12/2006	Braun et al.	N/A	N/A
7209790	12/2006	Thompson et al.	N/A	N/A
7218231	12/2006	Higham	N/A	N/A
7225241	12/2006	Yada	N/A	N/A
7230536	12/2006	Shinada et al.	N/A	N/A
7242306	12/2006	Wildman et al.	N/A	N/A
7242308	12/2006	Ulrich et al.	N/A	N/A
7248880	12/2006	Gheorghiu et al.	N/A	N/A
7248933	12/2006	Wildman	N/A	N/A
7259676	12/2006	Knadle, Jr. et al.	N/A	N/A
7265668	12/2006	Brosius	N/A	N/A
7269427	12/2006	Hector et al.	N/A	N/A
7277048	12/2006	Hessing	N/A	N/A
7277889	12/2006	Addonisio et al.	N/A	N/A
7283046	12/2006	Culpepper et al.	N/A	N/A
7283423	12/2006	Holm et al.	N/A	N/A
7286868	12/2006	Govari	N/A	N/A
7292151	12/2006	Ferguson et al.	N/A	N/A
7295115	12/2006	Aljadeff et al.	N/A	N/A
7295132	12/2006	Steiner	N/A	N/A
7298359	12/2006	Kim et al.	N/A	N/A
7301451	12/2006	Hastings	N/A	N/A
7304577	12/2006	Waldner et al.	N/A	N/A
7304578	12/2006	Sayers et al.	N/A	N/A
7304579	12/2006	Diorio et al.	N/A	N/A
7307522	12/2006	Dawson	N/A	N/A
7313403	12/2006	Gong et al.	N/A	N/A
7315248	12/2007	Egbert	N/A	N/A
7315281	12/2007	Dejanovic et al.	N/A	N/A
7319386	12/2007	Collins, Jr. et al.	N/A	N/A
7319395	12/2007	Puzio et al.	N/A	N/A
7319397	12/2007	Chung et al.	N/A	N/A
7319412	12/2007	Coppinger et al.	N/A	N/A
7321305	12/2007	Göllü	N/A	N/A
7333018	12/2007	Singh et al.	N/A	N/A
7336187	12/2007	Hubbard, Jr. et al.	N/A	N/A
7336563	12/2007	Holm	N/A	N/A
7339477	12/2007	Puzio et al.	N/A	N/A
7339479	12/2007	Nishimura	N/A	N/A
7362656	12/2007	Holm	N/A	N/A
7370808	12/2007	Eastin	N/A	N/A
7375648	12/2007	Mulka et al.	N/A	N/A
7375654	12/2007	Culpepper et al.	N/A	N/A
7376123	12/2007	Reuss	N/A	N/A

7394385	12/2007	Franco, Jr. et al.	N/A	N/A
7411509	12/2007	Rosenfeld et al.	N/A	N/A
7415212	12/2007	Matsushita et al.	N/A	N/A
7417544	12/2007	Artem et al.	N/A	N/A
7443299	12/2007	Forster	N/A	N/A
7446664	12/2007	White	N/A	N/A
7473097	12/2008	Raby et al.	N/A	N/A
7474224	12/2008	Long et al.	N/A	N/A
7515044	12/2008	Welch et al.	N/A	N/A
7518502	12/2008	Austin et al.	N/A	N/A
7538679	12/2008	Shanks	N/A	N/A
7541927	12/2008	Dupler et al.	N/A	N/A
7551083	12/2008	Modes et al.	N/A	N/A
7567794	12/2008	Dempsey	N/A	N/A
7598853	12/2008	Becker et al.	N/A	N/A
7623250	12/2008	Moctezuma de la Barrera et al.	N/A	N/A
7627334	12/2008	Cohen et al.	N/A	N/A
7664686	12/2009	Czyszczewski et al.	N/A	N/A
7667572	12/2009	Husak et al.	N/A	N/A
RE41236	12/2009	Seely	N/A	N/A
7714728	12/2009	Koblasz	N/A	N/A
7724147	12/2009	Brown	N/A	N/A
7725195	12/2009	Lima et al.	N/A	N/A
7734476	12/2009	Wildman et al.	N/A	N/A
7737850	12/2009	Malik	N/A	N/A
7750793	12/2009	Juels	N/A	N/A
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7765286	12/2009	Mark	N/A	N/A
7800914	12/2009	Dully	N/A	N/A
7844505	12/2009	Arneson et al.	N/A	N/A
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7869861	12/2010	Moctezuma de la Barrera et al.	N/A	N/A
7890349	12/2010	Cole et al.	N/A	N/A
7893842	12/2010	Deutsch	N/A	N/A
7907053	12/2010	Wildman et al.	N/A	N/A
7916023	12/2010	Rado	N/A	N/A
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7933780	12/2010	De La Hueraga	N/A	N/A
7958087	12/2010	Blumenau	N/A	N/A
7966008	12/2010	Graves et al.	N/A	N/A
8026821	12/2010	Reeder et al.	N/A	N/A
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8102254	12/2011	Becker et al.	N/A	N/A
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8190730	12/2011	Dempsey	N/A	N/A
8223009	12/2011	Anderson et al.	N/A	N/A
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8284047	12/2011	Collins, Jr. et al.	N/A	N/A
8310364	12/2011	Derks et al.	N/A	N/A
8319633	12/2011	Becker et al.	N/A	N/A
8321302	12/2011	Bauer et al.	N/A	N/A
8334898	12/2011	Ryan et al.	N/A	N/A
8334901	12/2011	Ganick et al.	N/A	N/A
8390462	12/2012	Belz et al.	N/A	N/A
8416072	12/2012	Tenarvitz	N/A	N/A
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8447626	12/2012	Sun et al.	N/A	N/A
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8461968	12/2012	Ball et al.	N/A	N/A
8461982	12/2012	Becker et al.	N/A	N/A
8514071	12/2012	Derks et al.	N/A	N/A
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Background/Summary

(1) The present application is a continuation of U.S. application Ser. No. 16/930,427, filed Jul. 16, 2020, now U.S. Patent No. 11,911,325, which is a continuation-in-part of U.S. application Ser. No. 16/743,340, filed Jan. 15, 2020, which claims the benefit, under 35 U.S.C. § 119(e), of U.S. Provisional Application No. 62/810,445, filed Feb. 26, 2019, each of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

(1) The present disclosure relates to patient beds and particularly, to patient beds having equipment used in connection with determining locations of the patient beds in a healthcare facility. More particularly, the present disclosure relates to patient beds having bed interfaces for manual location of the patient beds.

(2) Some healthcare facilities, such as hospitals, nursing homes, and the like, have nurse call systems that receive bed status data via a wired connection to a patient bed. For example, the NaviCare® Nurse Call system available from Hill-Rom Company, Inc. uses a 37-pin cable to interconnect patient beds to a bed interface unit (BIU) or network interface unit (NIU) or audio station bed connector (ASBC). The BTU's, NIU's, and ASBC's have identification (ID) codes such as serial numbers and/or MAC addresses that can be correlated with a room location in the healthcare facility. Thus, by connecting the patient beds to the respective BTU's, NIU's, or ASBC's, as the case may be, a remote computer device such as a server of the nurse call system is able to receive bed ID data and location ID data and determine the room locations of the various beds in the healthcare facility.

(3) However, not all healthcare facilities have nurse call systems to which patient beds couple via a wired connection. In recent times, some patient beds are equipped with wireless communication circuitry for WiFi communication between the patient beds and wireless access points (WAP's) of the healthcare facility. The radio frequency (RF) signals from the WiFi circuitry of the patient beds is able to pass through walls, floors, and ceilings such that multiple WAP's may receive the RF signals transmitted from the beds. Accordingly, the room locations of such beds cannot be determined with absolute certainty based on which WAP's are able to communicate with the beds. What is needed, therefore, is an inexpensive way to determine the locations of patient beds having wireless communication capability but that are not coupled to nurse call systems via wired connections.

SUMMARY

(4) An apparatus, system or method may comprise one or more of the features recited in the appended claims and/or the following features which, alone or in any combination, may comprise patentable subject matter:

(5) According to a first aspect of the present disclosure, a patient bed may include a frame that may be configured to support a patient, circuitry that may be carried by the frame, and a graphical user interface (GUI) that may be carried by the frame and that may be coupled to the circuitry. The graphical user interface may display at least one user interface screen that may be used by a caregiver to manually enter location data that may be indicative of a location in a healthcare facility

at which the patient bed may be located. The circuitry may be configured to transmit the location data entered by the caregiver and a bed identification (ID) from the bed. The circuitry may command the GUI to display a first screen of the at least one interface screen that may be used by the caregiver to manually enter location data. The first screen may be displayed after a threshold period of time may have elapsed subsequent to a power plug of the circuitry being plugged into an alternating current (AC) outlet of the healthcare facility and subsequent to casters of the frame being braked.

(6) In some embodiments of the first aspect, the location data may include a room number of the healthcare facility. If desired, the GUI of the first aspect may include a change location button that may be selectable by the caregiver to initiate a change of the location data prior to the threshold period of time elapsing. A room menu screen may appear on the GUI in response to the change location button being selected. The room menu screen may permit the caregiver to select a room number from a list of room numbers. Alternatively or additionally, a campus icon and a unit icon may appear on the GUI in response to the change location button being selected. The campus icon may be selectable to cause a campus menu list to be displayed on the GUI and the unit icon may be selectable to cause a unit menu list to be displayed on the GUI.

(7) The first screen appearing on the GUI after the threshold period of time may have elapsed may include a button that may be selectable to initiate manual entry of the location data. This button on the first screen is different than the change location button mentioned above. However, a room menu screen may appear on the GUI in response to the button being selected on the first screen. The room menu screen may permit the caregiver to select a room number from a list of room numbers. Alternatively or additionally, a campus icon and a unit icon may appear on the GUI in response to the button being selected on the first screen. The campus icon may be selectable to cause a campus menu list to be displayed on the GUI and the unit icon may be selectable to cause a unit menu list to be displayed on the GUI.

(8) In some embodiments of the first aspect, the circuitry may include a wireless communication module that may be configured to wirelessly transmit the location data and the bed ID to a wireless access point for delivery to at least one remote computer for purposes of making a bed-to-room association. Optionally, the wireless communication module also may be configured to transmit bed status data from the bed. If desired, the circuitry may be configured to play a voice prompt to remind the caregiver to manually enter the location data after the threshold period of time may have elapsed.

(9) According to a second aspect of the present disclosure, a patient bed may include a frame that may be configured to support a patient, circuitry that may be carried by the frame, and a graphical user interface (GUI) that may be carried by the frame and that may be coupled to the circuitry. The graphical user interface may display at least one user interface screen that may be used by a caregiver to manually enter location data that may be indicative of a location in a healthcare facility at which the patient bed may be located. The circuitry may be configured to transmit the location data entered by the caregiver and a bed identification (ID) from the bed. The circuitry may be configured to play a voice prompt to remind the caregiver to manually enter the location data after a threshold period of time may have elapsed subsequent to a power plug of the circuitry being plugged into an alternating current (AC) outlet of the healthcare facility and subsequent to casters of the frame being braked.

(10) In some embodiments of the second aspect, the circuitry also may command the GUI to display a first screen of at least one interface screen that may be used by the caregiver to manually enter location data after the threshold period of time may have elapsed. The location data may include a room number, for example. Optionally, the first screen appearing on the GUI may include a button that is selectable to initiate manual entry of the location data. If desired, a room menu screen may appear on the GUI in response to the button being selected on the first screen. The room menu screen may permit the caregiver to select a room number from a list of room numbers.

Alternatively or additionally, a campus icon and a unit icon may appear on the GUI in response to the button being selected on the first screen. The campus icon may be selectable to cause a campus menu list to be displayed on the GUI and the unit icon may be selectable to cause a unit menu list to be displayed on the GUI.

(11) Further according to the second aspect, the circuitry may include a wireless communication module that may be configured to wirelessly transmit the location data and the bed ID to a wireless access point for delivery to at least one remote computer for purposes of making a bed-to-room association. Optionally, the wireless communication module also may transmit bed status data from the bed.

(12) In some embodiments of the second aspect, the GUI may include a change location button that may be selectable by the caregiver to initiate a change of the location data prior to the threshold period of time elapsing. A room menu screen may appear on the GUI in response to the change location button being selected. The room menu screen may permit the caregiver to select a room number from a list of room numbers. Alternatively or additionally, a campus icon and a unit icon may appear on the GUI in response to the change location button being selected. The campus icon may be selectable to cause a campus menu list to be displayed on the GUI and the unit icon may be selectable to cause a unit menu list to be displayed on the GUI.

(13) According to a third aspect of the present disclosure, a method of manually associating a patient bed to a location in a healthcare facility may be provided. The method may include determining with circuitry of the patient bed that a power plug of the patient bed may be plugged into an alternating current (AC) outlet of the healthcare facility. The method may also include determining with the circuitry of the patient bed that casters of the patient bed may be braked. Subsequent to determining that the power plug may be plugged into the AC outlet and subsequent to determining that the casters may be braked, the method may further include determining with the circuitry of the patient bed whether a threshold period of time may have elapsed. After the threshold period of time may have elapsed, the method may include displaying on a graphical user interface (GUI) of the patient bed at least one user interface screen that may be configured to be used by a caregiver to manually enter location data that may be indicative of a location in the healthcare facility at which the patient bed may be located. The method further may include transmitting from the patient bed using the circuitry of the patient bed the location data entered by the caregiver and a bed identification (ID).

(14) In some embodiments of the third aspect, the method further may include displaying on the GUI a change location button that may be selectable by the caregiver to initiate a change of the location data prior to the threshold period of time elapsing. Optionally, in response to selection of the change location button by the caregiver, the method may include displaying a room menu screen on the GUI. The room menu screen may be configured to permit the caregiver to select a room number from a list of room numbers. Alternatively or additionally, in response to selection of the change location button by the caregiver, the method may include displaying a campus icon and a unit icon on the GUI. The campus icon may be selectable to cause a campus menu list to be displayed on the GUI and the unit icon may be selectable to cause a unit menu list to be displayed on the GUI.

(15) Further according to the third aspect, displaying on the graphical user interface (GUI) of the patient bed at least one user interface screen may include displaying a first screen. The first screen may appear on the GUI after the threshold period of time may have elapsed and may include a button that may be selectable to initiate manual entry of the location data. The button on the first screen is different than the change location button mentioned above. The method of the third aspect may further include, in response to selection of the button by the caregiver on the first screen, displaying a room menu screen on the GUI. The room menu screen may be configured to permit the caregiver to select a room number from a list of room numbers. Alternatively or additionally, in response to selection of the button by the caregiver on the first screen, the method of the third

aspect may include displaying a campus icon and a unit icon on the GUI. The campus icon may be selectable to cause a campus menu list to be displayed on the GUI and the unit icon may be selectable to cause a unit menu list to be displayed on the GUI.

(16) If desired, transmitting from the patient bed using the circuitry of the patient bed the location data entered by the caregiver and the bed ID may include using a wireless communication module to wirelessly transmit the location data and the bed ID to a wireless access point for delivery to at least one remote computer for purposes of making a bed-to-room association. Optionally, the method of the third aspect may further include using the wireless communication module to wirelessly transmit bed status data from the bed. Further optionally, the method of the third aspect may include playing a voice prompt using the circuitry of the patient bed to remind the caregiver to manually enter the location data after the threshold period of time may have elapsed.

(17) According to a fourth aspect of the present disclosure, a method of manually associating a patient bed to a location in a healthcare facility may be provided. The method may include determining with circuitry of the patient bed that a power plug of the patient bed may be plugged into an alternating current (AC) outlet of the healthcare facility. The method may also include determining with the circuitry of the patient bed that casters of the patient bed may be braked. Subsequent to determining that the power plug may be plugged into the AC outlet and subsequent to determining that the casters may be braked, the method may further include determining with the circuitry of the patient bed whether a threshold period of time may have elapsed. After the threshold period of time may have elapsed, the method may include using the circuitry to play a voice prompt to remind a caregiver to manually enter location data using a graphical user interface (GUI) of the patient bed that may display at least one user interface screen configured to be used by the caregiver to manually enter the location data that may be indicative of a location in the healthcare facility at which the patient bed may be located. The method further may include transmitting from the patient bed using the circuitry of the patient bed the location data entered by the caregiver and a bed identification (ID).

(18) In some embodiments of the fourth aspect, the at least one user interface screen may include a first screen and the method may include displaying on the first screen a button that may be selectable to initiate manual entry of the location data. If desired, in response to selection of the button by the caregiver on the first screen, the method may include displaying a room menu screen on the GUI. The room menu screen may be configured to permit the caregiver to select a room number from a list of room numbers. Alternatively or additionally, in response to selection of the button by the caregiver on the first screen, the method of the fourth aspect may include displaying a campus icon and a unit icon on the GUI. The campus icon being selectable to cause a campus menu list to be displayed on the GUI and the unit icon may be selectable to cause a unit menu list to be displayed on the GUI.

(19) If desired, transmitting from the patient bed using the circuitry of the patient bed the location data entered by the caregiver and the bed ID may include using a wireless communication module to wirelessly transmit the location data and the bed ID to a wireless access point for delivery to at least one remote computer for purposes of making a bed-to-room association. Optionally, the method of the fourth aspect may further include using the wireless communication module to wirelessly transmit bed status data from the bed.

(20) Further according to the fourth aspect, the method may include displaying on the GUI a change location button that may be selectable by the caregiver to initiate a change of the location data prior to the threshold period of time elapsing. Optionally, in response to selection of the change location button by the caregiver, the method may include displaying a room menu screen on the GUI. The room menu screen may be configured to permit the caregiver to select a room number from a list of room numbers. Alternatively or additionally, in response to selection of the change location button by the caregiver, the method may include displaying a campus icon and a unit icon on the GUI. The campus icon may be selectable to cause a campus menu list to be displayed on the

GUI and the unit icon may be selectable to cause a unit menu list to be displayed on the GUI.

(21) According to a fifth aspect of the disclosed embodiments, a patient bed may include a frame configured to support a patient. Circuitry may be carried by the frame. A graphical user interface (GUI) may be carried by the frame and coupled to the circuitry. The graphical user interface may display at least one user interface screen that is used by a caregiver to manually validate patient identification data indicative of a identification of a patient positioned on the patient bed. The circuitry may be configured to transmit the patient identification data validated by the caregiver and a bed identification (ID) from the bed. The circuitry may command the GUI to display a first screen of the at least one interface screen that is used by the caregiver to validate the patient identification data. The circuitry may command the GUI to display a second screen of the at least one interface screen that displays a patient validation icon indicative of whether the patient identification data has been validated.

(22) In some embodiments of the fifth aspect, a weigh scale may be configured to output a signal to the circuitry indicative of a weight of the patient. The circuitry may command the GUI to display a third screen displaying the weight of the patient. The third screen may include a save icon.

Selection of the save icon may cause the circuitry to transmit data indicative of the weight of the patient and the validated patient identification data to an electronic medical record if the patient identification data has been validated. Selection of the save icon may cause the circuitry to transmit the data indicative of the weight of the patient and invalidated patient identification data to an electronic medical record if the patient identification data has not been validated.

(23) If desired, in the fifth aspect, the circuitry may be configured to prompt the caregiver to manually enter location data of the patient bed after a threshold period of time elapses subsequent to a power plug of the circuitry being plugged into an outlet of a healthcare facility and subsequent to casters of the frame being braked. The location data may include a room number of the healthcare facility.

(24) Optionally, in the fifth aspect, the validation icon may be illuminated in a first color in response to the patient identification data being validated. The validation icon may be illuminated in a second color that is different from the first color in response to the patient identification data not being validated. The second screen of the at least one interface screen may display a name of the patient in response to the patient identification data being validated.

(25) Further, in the fifth aspect, the patient identification data may be temporarily invalidated in response to the circuitry being unplugged from an outlet of a healthcare facility. The patient identification data may be temporarily invalidated in response to the patient exiting the bed. The patient identification data may be invalidated in response to the patient bed being moved within a healthcare facility. The patient identification data may be invalidated in response to a new patient being positioned on the patient bed.

(26) It may be contemplated that, in the fifth aspect, the first screen appearing on the GUI after the threshold period of time elapses includes a button that is selectable to initiate manual validation of the patient identification data. The circuitry may include a wireless communication module configured to wirelessly transmit the patient identification data to a wireless access point for delivery to at least one remote computer for purposes of making a bed-to-patient association. The wireless communication module may be configured to transmit bed status data from the bed. The circuitry may be configured to play a voice prompt to remind the caregiver to manually validate the patient identification data after the threshold period of time elapses.

(27) According to a sixth aspect of the disclosed embodiments, a patient bed may include a frame configured to support a patient. Circuitry may be carried by the frame. A graphical user interface (GUI) may be carried by the frame and coupled to the circuitry. The graphical user interface may display at least one user interface screen that is used by a caregiver to manually validate patient identification data indicative of a patient positioned on the patient bed. The circuitry may be configured to transmit the patient identification data validated by the caregiver and a bed

identification (ID) from the bed. The circuitry may command the GUI to display a first screen of the at least one interface screen that is used by the caregiver to validate the patient identification data. The circuitry may command the GUI to display a second screen of the at least one interface screen that displays a patient validation icon indicative of whether the patient identification data has been validated. A weigh scale may be configured to output a signal to the circuitry indicative of a weight of the patient. The circuitry may command the GUI to display a third screen displaying the weight of the patient. The circuitry may be configured to prompt the caregiver to manually enter a location data of the patient bed. The location data may be indicative of a location of the patient bed in a healthcare facility.

(28) In some embodiments of the sixth aspect, the third screen may include a save icon. Selection of the save icon may cause the circuitry to transmit data indicative of the weight of the patient and the validated patient identification data to an electronic medical record if the patient identification data has been validated. Selection of the save icon may cause the circuitry to transmit the data indicative of the weight of the patient and invalidated patient identification data to an electronic medical record if the patient identification data has not been validated. The location data may include a room number of the healthcare facility.

(29) Optionally, in the sixth aspect, the validation icon may be illuminated in a first color in response to the patient identification data being validated. The validation icon may be illuminated in a second color that is different from the first color in response to the patient identification data not being validated.

(30) If desired, in the sixth aspect, the second screen of the at least one interface screen may display a name of the patient in response to the patient identification data being validated. The patient identification data may be temporarily invalidated in response to at least one of the circuitry being unplugged from an outlet of the healthcare facility or the patient exiting the bed. The patient identification data may be invalidated in response to at least one of the patient bed being moved within the healthcare facility or a new patient being positioned on the patient bed.

(31) Additional features, which alone or in combination with any other feature(s), such as those listed above and/or those listed in the claims, may comprise patentable subject matter and will become apparent to those skilled in the art upon consideration of the following detailed description of various embodiments exemplifying the best mode of carrying out the embodiments as presently perceived.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description particularly refers to the accompanying figures, in which:

(2) FIG. 1 is a diagrammatic view of a bed manual locating system showing a patient bed wirelessly coupled to a network via a wireless access point for communication with a real time locating system (RTLS) server, an electronic medical record (EMR) server, and a communications server;

(3) FIG. 2 is a block diagram showing various components of the patient bed of FIG. 1 and also showing the RTLS, EMR, nurse call, and communications systems of the network;

(4) FIG. 3 is a diagrammatic view showing a graphical user interface (GUI) of the patient bed having a touch screen display that displays a window having a room number and an edit icon or button;

(5) FIGS. 4A and 4B together make up a diagrammatic view having a flow chart portion in FIG. 4A and having screen flows showing how the GUI of the bed is used to manually enter location data for the patient bed;

(6) FIGS. 5A-5C together make up a diagrammatic view having screen flows showing how the

- GUI of the bed is used to manually enter patient identification data for the patient bed;
- (7) FIG. 6 is a diagrammatic view having screen flows showing how the GUI of the bed is used to enter patient weight data for the patient bed;
- (8) FIG. 7 is a diagrammatic view having screen flows showing how the GUI of the bed is used to enter patient identification data for the patient bed;
- (9) FIG. 8 is a diagrammatic view of a screen showing how the GUI of the bed is used to adjust settings; and
- (10) FIG. 9 is a diagrammatic view of a revalidation screen for revalidating location information.
- DETAILED DESCRIPTION**

(11) A bed manual locating system **10** includes a plurality of patient beds **12**, only one of which is shown in FIG. 1, coupled to a network **18** of a healthcare facility such as a hospital, outpatient care facility, nursing home, and the like. In the illustrative example, patient bed **12** is coupled to network **18** via a wireless access point (WAP) **14** which may or may not be located in the respective patient room with the bed **12**. It should be understood that, although only one WAP **14** is shown in FIG. 1, multiple WAP's **14** may be in wireless communication with bed **12**. Illustratively, bed **12** includes a wireless communication module **16**, as shown in FIG. 2, that sends wireless signals, such as radio frequency (RF) signals, or transmissions to WAP **14** and that receives wireless signals, such as RF signals, or transmissions from WAP. Thus, wireless communication module **16** includes a transceiver in some embodiments.

(12) As shown diagrammatically in FIGS. 1 and 2, a communication server **20** of a corresponding communication system **21**, a nurse call server **22** of a corresponding nurse call system **23**, a real time locating system (RTLS) server **24** of a corresponding RTLS **25**, and an electronic medical records (EMR) server **26** of an EMR system **27** are each coupled to network **18**. Block **27** also is intended to represent other remote computers of network **18** that are included in system **10** in lieu of or in addition to depicted servers **20**, **22**, **24**, **26**. For example, system **10** includes an admission/discharge/transfer (ADT) server and/or a bed status data server in some embodiments. Double headed arrows **29** in FIGS. 1 and 2 represent the bidirectional communication links between network **18**, WAP **14** (and the corresponding bed **12**) and each of servers **20**, **22**, **24**, **27** of respective systems **21**, **23**, **25**, **27** and therefore, with each other. Communication links **29** include wired communication links or wireless communication links or both at the option of the designer of system **10** in any given healthcare facility. The wireless communication between the WAP's **14** of network **18** and module **16** of bed **12** contemplated by this disclosure includes Bluetooth (BT), Bluetooth Low Energy (BLE), Zigbee, Z-Wave, and WiFi (e.g., any of the 802.11.sub.x protocols). However, this is not rule out other types of wireless communication between bed **12** and an appropriately configured transceiver in addition to or in lieu of WAP **14**, including infrared (IR) communications, ultrasonic (US) communications, ultra-wideband (UWB) communications, and so forth.

(13) The RTLS **25** of system **10** includes wireless transceiver units **28** placed throughout the healthcare facility. Only one such unit **28** is depicted diagrammatically in FIG. 1. The RTLS **25** of system **10** also includes caregiver locating or tracking tags or badges **30** that are worn by caregivers. Each of the transceiver units **28** receives a wireless signal from the badges **30** of each of the caregivers wearing badges **30** and that are within communication range of the respective unit **28** as indicated diagrammatically by arrow **32** in FIG. 1. The wireless signal from each badge **30** includes badge identification data (ID) which is unique to the corresponding badge **30**. Unit **28** then transmits its ID, which corresponds to a particular location in the healthcare facility, and the badge (ID) to RTLS server **24** as indicated diagrammatically by a bidirectional communication link **34** in FIG. 1. Based on the received badge ID and the location ID from unit **28**, server **24** determines the location of the caregiver within the healthcare facility. Some healthcare facilities do not include RTLS **25** in which case server **24**, transceivers **28**, and badges **30** are omitted from system **10**.

(14) In some embodiments, RTLS server **24** is included in system **10** and transceivers **28** and

badges **30** are omitted. In such embodiments, RTLS server **24** serves as a data repository for bed-to-room associations that are established via manual entry at beds **12** as will be discussed in further detail below in connection with FIGS. **3**, **4A** and **4B**. Alternatively, the bed-to-room associations established via manual entry at beds **12** are stored in another server, such as nurse call server **22**, EMR server, or any other server of system **10**, including servers **20**. Thus, servers **20**, **22**, **26**, or other servers of system **10**, may serve multiple purposes and so these may be configured to serve the RTLS function along with their other function(s) (e.g., nurse call, EMR, communications, etc.).

(15) In some embodiments having RTLS **25** with transceivers **28** and tags **30** to track caregiver locations, the location of assets such as beds **12** can also be tracked in a similar manner by server **24** by attaching asset tags that are substantially the same as badges **30** to the assets to be tracked. However, in such systems **10** in which beds **12** have asset tags for tracking by RTLS **25**, then manual association of beds **12** to rooms as discussed below in connection with FIGS. **3**, **4A** and **4B** is not needed. However, hybrid systems **10** in which some locations of beds **12** are determined by signals from respective asset tags attached to the beds **12** and in which some locations of beds **12** are determined by manual entry at the respective beds **12**, are contemplated by the present disclosure. In FIG. **1**, the caregiver is shown carrying a tablet computer **38** which is configured to communicate wirelessly with other devices of network **18** such as via communications server **20**. Mobile phones are carried by caregivers in addition to, or in lieu of tablet computers **38**, and are in communication with communication server **20** in some embodiments.

(16) As alluded to above, the present disclosure is primarily focused on manual entry of locating information at beds **12**. However, a discussion is provided below of the basic components and operation of various features of bed **12** so that an understanding of the types of bed status data transmitted wirelessly by module **16** to WAP **14** can be gained. From the below discussion it will be appreciated that beds **12** having capability for manual entry of location information are by themselves, or in combination with the other components of system **10**, a practical application of the concepts and ideas disclosed herein. Bed **12** includes a patient support structure such as a frame **40** that supports a surface or mattress **42** as shown in FIG. **1**. It should be understood that FIG. **1** shows some details of one possible bed **12** having one specific configuration. In particular, illustrative bed **12** of FIG. **1** is the CENTRELLA® bed available from Hill-Rom Company, Inc. Other aspects of illustrative bed **12** are shown and described in more detail in U.S. Patent Application Publication No. 2018/0161225 A1 which is hereby expressly incorporated by reference herein to the extent not inconsistent with the present disclosure which shall control as to any inconsistencies. However, this disclosure is applicable to other types of patient support apparatuses **12** having other configurations, including other types of beds, surgical tables, examination tables, stretchers, chairs, wheelchairs, patient lifts and the like.

(17) Still referring to FIG. **1**, frame **40** of bed **12** includes a base frame **48** (sometimes just referred herein to as a base **48**), an upper frame assembly **50** and a lift system **52** coupling upper frame assembly **50** to base **48**. Lift system **52** is operable to raise, lower, and tilt upper frame assembly **50** relative to base **48**. Bed **12** has a head end **54** and a foot end **56**. Patient bed **12** further includes a footboard **55** at the foot end **56** and a headboard **57** at the head end **54**. Illustrative bed **12** includes a pair of push handles **47** coupled to an upstanding portion **59** of base **48** at the head end **54** of bed **12**. Only a portion of one push handle **47** can be seen in FIG. **1**. Headboard **57** is coupled to upstanding portion **59** of base as well. Footboard **55** is coupled to upper frame assembly **50**. Base **48** includes wheels or casters **49** that roll along floor (not shown) as bed **12** is moved from one location to another. A set of foot pedals **51** are coupled to base **48** and are used to brake and release casters **49**.

(18) Illustrative patient bed **12** has four siderail assemblies coupled to upper frame assembly **50** as shown in FIG. **1**. The four siderail assemblies include a pair of head siderail assemblies **58** (sometimes referred to as head rails) and a pair of foot siderail assemblies **60** (sometimes referred to as foot rails). Each of the siderail assemblies **58** and **60** is movable between a raised position, as

shown in FIG. 1, and a lowered position (not shown). Siderail assemblies **58, 60** are sometimes referred to herein as siderails **58, 60**. Each siderail **58, 60** includes a barrier panel **64** and a linkage **66**. Each linkage **66** is coupled to the upper frame assembly **50** and is configured to guide the barrier panel **64** during movement of siderails **58, 60** between the respective raised and lowered positions. Barrier panel **64** is maintained by the linkage **66** in a substantially vertical orientation during movement of siderails **58, 60** between the respective raised and lowered positions.

(19) Upper frame assembly **50** includes various frame elements **68**, shown in FIG. 1, that form, for example, a lift frame and a weigh frame supported with respect to the lift frame by a set of load cells **72** of a scale and/or bed exit/patient position monitoring (PPM) system **70** of bed **12**, as shown diagrammatically in FIG. 2. A patient support deck **74**, shown diagrammatically in FIG. 2, is carried by the weigh frame portion of upper frame assembly **50** and supports mattress **42** thereon. Data relating to the operation of the scale and/or bed exit/PPM system **70** is among the features of bed **12** for which bed status data is transmitted wirelessly from module **16** to one or more WAP's **14**.

(20) Patient support deck **74** includes a head section **80**, a seat section **82**, a thigh section **83** and a foot section **84** in the illustrative example as shown diagrammatically in FIG. 2. Sections **80, 83, 84** are each movable relative to the weigh frame portion of upper frame assembly **50**. For example, head section **80** pivotably raises and lowers relative to seat section **82** whereas foot section **84** pivotably raises and lowers relative to thigh section **83**. Additionally, thigh section **83** articulates relative to seat section **82**. Also, in some embodiments, foot section **84** is extendable and retractable to change the overall length of foot section **84** and therefore, to change the overall length of deck **74**. For example, foot section **84** includes a main portion **85** and an extension **87** in some embodiments as shown diagrammatically in FIG. 2.

(21) In the illustrative embodiment, seat section **82** is fixed in position with respect to the weigh frame portion of upper frame assembly **50** as patient support deck **74** moves between its various patient supporting positions including a horizontal position to support the patient in a supine position, for example, and a chair position (not shown) to support the patient in a sitting up position. In other embodiments, seat section **82** also moves relative to upper frame assembly **50**, such as by pivoting and/or translating. Of course, in those embodiments in which seat section **82** translates relative to the upper frame assembly **50**, the thigh and foot sections **83, 84** also translate along with seat section **82**. As bed **12** moves from the horizontal position to the chair position, foot section **84** lowers relative to thigh section **83** and shortens in length due to retraction of the extension **87** relative to main portion **85**. As bed **12** moves from the chair position to the horizontal position, foot section **84** raises relative to thigh section **83** and increases in length due to extension of the extension **87** relative to main portion **85**. Thus, in the chair position, head section **80** extends upwardly from upper frame assembly **50** and foot section **84** extends downwardly from thigh section **83**.

(22) As shown diagrammatically in FIG. 2, bed **12** includes a head motor or actuator **90** coupled to head section **80**, a knee motor or actuator **92** coupled to thigh section **83**, a foot motor or actuator **94** coupled to foot section **84**, and a foot extension motor or actuator **96** coupled to foot extension **87**. Motors **90, 92, 94, 96** may include, for example, an electric motor of a linear actuator. In those embodiments in which seat section **82** translates along upper frame assembly **50** as mentioned above, a seat motor or actuator (not shown) is also provided. Head motor **90** is operable to raise and lower head section **80**, knee motor **92** is operable to articulate thigh section **83** relative to seat section **82**, foot motor **94** is operable to raise and lower foot section **84** relative to thigh section **83**, and foot extension motor **96** is operable to extend and retract extension **87** of foot section **84** relative to main portion **85** of foot section **84**. Data relating to the operation of motors **90, 92, 94, 96** and the positions of deck sections **80, 82, 83, 84** is among the features of bed **12** for which bed status data is transmitted wirelessly from module **16** to one or more WAP's **14**.

(23) In some embodiments, bed **12** includes a pneumatic system **98** that controls inflation and

deflation of various air bladders or cells of mattress **42**. The pneumatic system **98** is represented in FIG. **2** as a single block but that block **98** is intended to represent one or more air sources (e.g., a fan, a blower, a compressor) and associated valves, manifolds, air passages, air lines or tubes, pressure sensors, and the like, as well as the associated electric circuitry, that are typically included in a pneumatic system for inflating and deflating air bladders of mattresses. Operation of pneumatic system **98** is among the features of bed **12** for which bed status data is transmitted wirelessly from module **16** to one or more WAP's **14**.

(24) As also shown diagrammatically in FIG. **2**, lift system **52** of bed **12** includes one or more elevation system motors or actuators **100**, which in some embodiments, comprise linear actuators with electric motors. Thus, actuators **100** are sometimes referred to herein as motors **100** and operation of the motors **100** is among the features of bed **12** for which bed status data is transmitted wirelessly from module **16** to one or more WAP's. Alternative actuators or motors contemplated by this disclosure include hydraulic cylinders and pneumatic cylinders, for example. The motors **100** of lift system **52** are operable to raise, lower, and tilt upper frame assembly **50** relative to base **48**. In the illustrative embodiment, one of motors **100** is coupled to, and acts upon, a set of head end lift arms **102** and another of motors **100** is coupled to, and acts upon, a set of foot end lift arms **104** to accomplish the raising, lowering and tilting functions of upper frame **50** relative to base **48**. Guide links **105** are coupled to base **48** and to lift arms **104** in the illustrative example as shown in FIG. **1**.

(25) Each of siderails **58** includes a first user control panel **106** coupled to the outward side of the associated barrier panel **64**. Controls panels **106** include various buttons that are used by a caregiver to control associated functions of bed **12**. For example, control panel **106** includes buttons that are used to operate head motor **90** to raise and lower the head section **80**, buttons that are used to operate knee motor **92** to raise and lower the thigh section **83**, and buttons that are used to operate motors **100** to raise, lower, and tilt upper frame assembly **50** relative to base **48**. In some embodiments, control panel **106** also includes buttons that are used to operate motor **94** to raise and lower foot section **84** and buttons that are used to operate motor **96** to extend and retract foot extension **87** relative to main portion **85**. Each of siderails **58** also includes a second user control panel **108** coupled to the inward side of the associated barrier panel **64**. Controls panels **108** include various buttons that are used by a patient to control associated functions of bed **12**. In some embodiments, the buttons of control panels **106**, **108** comprise membrane switches that are used to control head motor **90** and knee motor **92**.

(26) As shown diagrammatically in FIG. **2**, bed **12** includes control circuitry **110** that is electrically coupled to motors **90**, **92**, **94**, **96** and to motors **100** of lift system **52**. Control circuitry **110** is sometimes referred to as a "controller." Control circuitry **110** is represented diagrammatically as a single block in FIG. **2**, but control circuitry **110** in some embodiments, comprises various circuit boards, electronics modules, and the like that are electrically and communicatively interconnected. Control circuitry **110** includes one or more microprocessors **112** or microcontrollers that execute software to perform the various control functions and algorithms described herein. Thus, circuitry **110** also includes memory **114** for storing software, variables, calculated values, and the like as is well known in the art. Memory **112** comprises, for example, one or more flash memory banks such as one or more EEPROM's, EPROM's, and the like. In some embodiments, memory **112** is included on the same integrated circuit chip as microprocessor **112**.

(27) As shown diagrammatically in FIG. **2**, an "other user inputs" block represents the various user inputs such as buttons of control panels **106**, **108**, for example, that are used by the caregiver or patient to communicate input signals to control circuitry **110** of bed **12** to command the operation of the various motors **90**, **92**, **94**, **96**, **100** of bed **12**, as well as commanding the operation of other functions of bed **12**. Bed **12** includes at least one graphical user input (GUI) or display screen **120** coupled to a respective siderail **58** as shown in FIG. **1**. Display screen **120** is coupled to control circuitry **110** as shown diagrammatically in FIG. **2**. In some embodiments, two graphical user interfaces **120** are provided and are coupled to respective siderails **58**. Alternatively or additionally,

one or more graphical user interfaces are coupled to siderails **60** and/or to one or both of the headboard **57** and footboard **55** or some other portion of bed **12** such as a support arm assembly extending upwardly from base frame **48** or upper frame **50**.

(28) Still referring to FIG. 2, wireless communication module **16** is also coupled electrically to control circuitry **110** and is configured for wireless communication with network **18** and its associated devices without the use of any wired bed cable in the illustrative embodiment. The communication of data from module **16** is among the features of bed **12** controlled by the bed operating software of control circuitry **110**.

(29) Still referring to FIG. 2, bed **12** includes various sensors to sense the states or positions of various portions of bed **12**. In the illustrative example, bed **12** includes an angle sensor **118** coupled to head section **80** to sense an angle of head section elevation (sometimes referred to as the head-of-bed (HOB) angle). Angle sensor **118** includes an accelerometer (single-axis or multi-axis) in some embodiments. In such embodiments, the HOB angle is measured with respect to a horizontal reference axis and/or with respect to a vertical reference axis depending upon the orientation of the accelerometer relative to head section **80** and depending upon the type of accelerometer used. In other embodiments, angle sensor **118** includes a rotary potentiometer which measures the HOB angle between head section **90** and another portion of frame **40** such as one of frame members **68** of upper frame assembly **50**. In further embodiments, angle sensor **118** is included in head motor **90** and has an output that correlates to the HOB angle. Motor **90** may include, for example, a shaft encoder, a Hall effect sensor, a rotary potentiometer, or some other sensor which serves as angle sensor **118** of bed **12** in such embodiments. Similar such sensors are included in elevation system motors **100** in some embodiments and are used to determine the position of upper frame assembly **50** relative to base **48** such as the height of upper frame assembly **50** and/or amount of tilt of upper frame assembly **50** relative to base **48**.

(30) Bed **12** also includes siderail position sensors **122** to sense the position (e.g., raised and/or lowered) of each of siderails **58**, **60** and one or more caster braking sensors **124** to sense whether casters **49** are braked or released. In some embodiments, sensors **122**, **124** include limit switches that are engaged or disengaged by a linkage mechanism, such as linkage **66** in the case of siderails **58**, **60**, to produce output signals indicative of the position of the respective mechanical structure. Alternatively, Hall effect sensors may be used as some or all of sensors **122**, **124** in some embodiments. The foregoing types of sensors **122**, **124** are just a couple examples of suitable sensors and therefore, this disclosure is intended to cover all types of sensors that may be used as sensors **122**, **124**. Each of the sensors mentioned above, including sensors internal to motors **100** and sensors **118**, **122**, **124** are each coupled electrical to control circuitry **110** for analysis and/or processing. Thus, data from sensors **118**, **122**, **124** is used by the bed operating software in connection with the control and operation of various features of bed **12** and is among the features of bed **12** for which bed status data is transmitted wirelessly from module **16** to one or more WAP's **14**.

(31) As shown in FIG. 1, bed **12** includes four status or alert lights **126**, **128a**, **128b**, **130** at foot end **56** corresponding to various monitored features of bed **12**. In the illustrative embodiment, for example, bed **12** includes a siderail position light **126**, a bed exit/PPM disabled light **128a**, a bed exit/PPM enabled light **128b**, and a bed lowest position light **130**. PPM is an acronym for “patient position monitoring.” Alert lights **126**, **128a**, **128b**, **130** are coupled to a lateral frame member of extension **87** of foot section **84** and are situated beneath footboard **55**. In other embodiments, alert lights **126**, **128a**, **128b**, **130** may be located elsewhere on bed **12** such as on base **48** and/or one or more of siderails **58**, **60**. In FIG. 2, alert lights **126**, **128a**, **128b**, **130** are represented diagrammatically as a single block and are coupled electrically to control circuitry **110** to control the manner in which alert lights **126**, **128a**, **128b**, **130** are illuminated as will be discussed in further detail below. In some embodiments, other alert lights **132**, shown diagrammatically as a single block in FIG. 2, are located elsewhere on bed **12**, such as on siderails **58**, **60**, and are illuminated to

convey information regarding other features of bed **12**, such as to indicate motor lockout conditions, alarm volume control levels, nurse call status, caster brake status, and the like.

(32) In some embodiments, alert lights **126**, **128b**, **130** are illuminated different colors to indicate certain statuses. For example, lights **126**, **128b**, **130** are illuminated a first color, such as green for example, if the associated bed condition is in an acceptable or satisfactory state. Lights **126**, **128b**, **130** are illuminated a second color, such as amber or yellow for example, if the associated bed condition is an undesirable or unsatisfactory state. Each of lights **126**, **128a**, **128b**, **130** has an icon on the lens of the respective light **126**, **128a**, **128b**, **130** corresponding to the monitored condition of bed **12**.

(33) In the illustrative example, if bed **12** has a falls risk protocol enabled (i.e., turned on) in which all of siderails **58**, **60** are required to be raised (or a subset of siderails **58**, **60** selected on GUI **120** is required to be raised), the light **126** is illuminated green if all of the siderails **58**, **60** (or selected subset of siderails **58**, **60**) are in the respective raised positions (e.g., the desirable or satisfactory condition) and the light **126** is illuminated amber, and in some embodiments flashed, if any one or more of siderails **58**, **60** (or selected subset of siderails **58**, **60**) is in the lowered position (e.g., the undesirable or unsatisfactory condition). In some embodiments, a lighted iconic image **126'** corresponding to the state of light **126** is projected onto the floor at the foot end of the bed **12** as shown in FIG. **1**. Image **126'** has the same color and icon as light **126**. If the falls risk protocol of bed **12** is disabled (i.e., turned off), then light **126** is turned off and no image **126'** is projected onto the floor by bed **12**.

(34) If the bed exit/PPM system of bed **12** is disabled (i.e., turned off), then light **128a** is illuminated blue and a corresponding blue lighted iconic image **128a'** is projected onto the floor by bed **12**. If the bed exit/PPM system of bed **12** is enabled (i.e., turned on), then light **128b** is illuminated and a corresponding lighted iconic image (not shown) is projected onto the floor by bed **12** and appears in the same general location as image **128a'**. Of course, when light **128b** is illuminated, light **128a** is turned off and image **128a'** is no longer projected onto the floor. Light **128b** and image **128b'** are illuminated green when the bed exit/PPM system is armed (aka enabled) and the patient is on the bed in the proper location (e.g., the desirable or satisfactory condition). Some embodiments of bed **12** have multiple modes (e.g., patient movement, pre-exit, and exiting modes) with varying levels of sensitivities at which an alarm condition is considered to exist. Light **128b** and image **128b'** are illuminated amber, and in some embodiments are flashed, if the bed exit/PPM system is armed and the patient is not properly positioned on bed, including being out of bed altogether (e.g., the undesirable or unsatisfactory condition).

(35) If bed **12** has the falls risk protocol enabled (i.e., turned on) control circuitry **110** monitors the position of the upper frame **50** relative to base frame **48** to assure that upper frame **50** is in its lowest position relative to base frame **48**. If upper frame **50** is in its lowest position (e.g., the desirable or satisfactory condition), the light **130** is illuminated green (e.g., the desirable or satisfactory condition). On the other hand, if upper frame **50** is not in its lowest position (e.g., the undesirable or unsatisfactory condition), the light **130** is illuminated amber, and in some embodiments flashed. In some embodiments, a lighted iconic image **130'** corresponding to the state of light **130** is projected onto the floor at the foot end of the bed **12** as shown in FIG. **1**. Image **130'** has the same color and icon as light **130**. If the falls risk protocol of bed **12** is disabled (i.e., turned off), then light **130** is turned off and no image **130'** is projected onto the floor by bed **12**.

(36) In some embodiments, an audible alarm of bed **12** may also sound under the control of control circuitry **110** if an unsatisfactory condition of a particular protocol or condition is detected. Lights **126**, **128b**, **130** are illuminated a third color if the associated protocol or condition is enabled for monitoring and at least one of the monitored bed statuses for the particular protocol or condition is undesirable (i.e., violated), but the associated alert has been suspended by the caregiver. If the alert has been suspended, any associated audible alarms may be turned off during the alarm suspension. A caregiver may suspend an alert associated with lights **126**, **128b**, **130**, for example, when

assisting a patient in getting out of bed **12** and going to the bathroom. The various alert conditions (aka alarm conditions) associated with the operation of alert lights **126**, **128a**, **126b**, **130** and the audible alarms, if any, of bed **12** is among the features of bed **12** for which bed status data is transmitted wirelessly from module **16** to one or more WAP's **14**.

(37) Referring now to FIG. **3**, a start-up screen **140** is shown on GUI **120**. GUI **120** is embodied as a touch screen display having various icons or buttons that are selectable by a caregiver to navigate to other screens for selection of various bed functions and entry of various types of data as will be discussed below, primarily in connection with manual entry of bed location data on GUI **120**. Start-up screen **140** appears on GUI, in some embodiments, in response to a caregiver (or really, any user for that matter) touching GUI **120** to wake GUI **120** from a dormant state in which GUI **120** is blacked out or, in some embodiments, in which GUI **120** displays a screen saver image of some sort.

(38) Illustrative start-up screen **140** has a bed location window **142** superimposed over a home screen **144**. Thus, FIG. **3** illustrates the situation in which a text string **146** of bed location information, illustratively “MED/SURG **1019-A**,” has been input manually during a previous use of GUI **120** by a caregiver. Window **142** includes an edit icon **148** that is selectable, such as by touching, to navigate to the screens that are used to edit the bed location information for whatever reason, such as if the bed **12** has been moved to a new room or if the hospital updates its room location naming convention. These edit screens are discussed below in connection with FIGS. **4A** and **4B**. Window **142** only appears over home screen **144** for a short period of time upon startup of GUI **120**, such as on the order of 5 to 30 seconds, and then disappears such that only home screen **144** remains displayed on GUI **120**. In some embodiments in which a healthcare facility is not equipped to receive information wirelessly from bed **12**, then the manual location entry feature using GUI **120** is disabled. In such situations, window **142** does not appear over home screen **144** upon startup of the GUI **120**.

(39) Home screen **144** includes a location icon **150** that is color coded to indicate whether bed location information or data has been entered manually using GUI **120** or, in some embodiments, if bed location data is otherwise transmitted to bed **12** either wirelessly or via a wired cable such as if bed **12** has an asset tag for locating as discussed above and RTLS server **24** sends a message to bed **12** regarding its location or if RTLS system **25** is otherwise configured to accurately determine the location of bed **12** without the use of an asset tag (e.g., the bed **12** is coupled via a cable to an NIU, BIU, or ASBC having a location ID indicative of room location). In other words, bed **12** is configured to permit bed location information to be entered manually using GUI **120** and to receive bed location information from equipment external to bed **12**, in some embodiments. In any event, if bed **12** has bed location information stored in memory **114** of control circuitry **110**, then bed location icon **150** is color coded green in some embodiments. If bed **12** does not have any bed location information stored in memory **114**, or if bed **12** has bed location information stored in memory **114** that is believed to be inaccurate based on sensed bed conditions such as unplugging of bed **12** from a wall outlet, unbraking or releasing of casters **49**, and/or movement of bed to a new location, then bed location icon **150** is color coded yellow or amber in some embodiments.

(40) For the sake of completeness, as shown in FIG. **3**, other illustrative icons and information displayed on home screen **144** includes, along a left side of home screen **144**, a head angle reading **152** corresponding to the HOB angle measured by angle sensor **118**, a bed exit/PPM icon **154** indicating the mode in which the scale/PPM system **70** is operating when enabled, and a urinary drainage bag lockout icon **156** indicating whether a urinary drainage bag (aka a Foley bag) is coupled to a bracket (not shown) attached to foot section **84** of bed **12**. A triangular edit button **155** is provided adjacent to icon **154** and is selectable to navigate to other screens for disabling the PPM system **70** of bed **12** or changing the mode in which the PPM system **70** operates. A similar triangular edit icon **157** is provided adjacent to icon **156** and is selectable to navigate to other screens for enabling and disabling the urinary drainage bag lockout feature of bed **12**. In this

regard, see U.S. Patent Application Publication No. 2018/0185222 A1 which is hereby incorporated by reference herein in its entirety to the extent not inconsistent with the present disclosure which shall control as to any inconsistencies.

(41) Home screen **144** includes an upper field **158** in which bed location icon **150** is shown. Upper field **158** also includes a help icon **160** that is selectable to navigate to various help screens of bed **12**, a service required icon **162** that is illuminated yellow or amber and flashes when bed service is needed, and a battery charge level indicator **164** having four segments that are illuminated green to indicate a battery charge level of a battery (not shown) of bed **12**. However, when the battery charge gets low, the segment to the left of indicator **164** turns yellow and flashes. The battery of bed **12** is used to supply power to various components of bed **12** when a power plug **166** at the end of a power cable **168** of bed **12** (see FIG. 2) is unplugged from an alternating current (AC) outlet **170** of the healthcare facility.

(42) Upper field **158** further includes a nurse call connectivity icon or indicator **172** that is color coded to indicate whether bed **12** is in communication with nurse call system **23** of the healthcare facility. In some embodiments, indicator **172** is color coded white when bed **12** is not in communication with nurse call system **23** which includes embodiments of system **10** in which nurse call system **23** is omitted altogether. In some embodiments, indicator is color coded green, such as by illuminating a green circle around the indicator **172** shown in FIG. 3. Finally, in the illustrative example, upper field **158** includes a WiFi connectivity icon or indicator **174** that is color coded to indicate whether wireless communication module **16** of bed **12** is communicating with one or more WAP's **14** and, in some embodiments, indicator **174** is a received signal strength indicator (RSSI) that indicates a relative level (e.g., low, medium, high) of the received signal strength of the wireless communication from the one or more WAP's received by module **16** of bed. In this regard, segments of indicator **174** are illuminated green when wireless communications exist between module **16** and at least one WAP **14** and indicator **174** is illuminated yellow or is not illuminated at all when no wireless communications exist between module **16** and WAP **14**.

(43) Still referring to FIG. 3, along a right side of home screen **144** a menu of icons are shown including a home button **176** that is selected to return to home screen **144** when viewing a screen on GUI other than home screen **144**, an alerts button **178** that is pressed to navigate to other screens for controlling alerting functions of bed **12**, a mattress control button **180** that is selected to navigate to other screens for controlling functions of mattress **42** and pneumatic system **98** of bed **12**, a scale control button **182** that is selected to navigate to other screens for controlling functions of scale system **70** of bed **12**, and a down arrow button **184** that is selected to cause other icons (aka buttons) to appear in the menu of icons at the right side of screen **144**. After button **184** is selected, an up arrow button (similar to button **184** but with the arrow pointing upwardly instead of downwardly) appears at the top of the menu of icons. In other words, the down arrow button **184** and up arrow button, when present, provide scrolling functionality to the menu of icons of screen **144**.

(44) Home screen **144** further includes a central field **186** that includes various control icons or buttons for controlling functions of bed **12** as shown in FIG. 3. Field **186** includes a foot extension button **188** that is selected to operate foot extension motor **96** to extend extension **87** relative to main portion **85** to lengthen foot section **84**. Field **186** also includes a foot retraction button **190** that is selected to operate motor **96** to retract extension **87** relative to main portion **85** to shorten foot section **84**. Above button **188**, field **186** includes a bed flat and level button **192** that operates one or more of motors **90, 92, 94**, as needed, to move deck sections **80, 82, 83, 84** into a substantially flat or substantially coplanar orientation relative to each other and to operate one or more of the elevation system motors **100**, as needed, to move the upper frame **50**, and therefore the deck **74**, into a substantially horizontal or substantially level orientation. The word “substantially” herein is intended to mean within plus or minus 10%, or less, of the desired condition and, at a minimum, within manufacturing tolerances of the desired condition. The movements of bed **12**

associated with buttons **188, 190, 192** occur only while the respective buttons **188, 190, 192** are pressed or touched. Thus, if the caregiver disengages from any of buttons **188, 190, 192** the associated bed movement stops.

(45) Field **186** further includes a stand assist button **194** situated above button **190** in the illustrative example. Button **194** is selected when a patient is sitting at a side of the mattress **42** of bed **12** and is getting ready to stand up from the bed. While button **194** is touched, motor **90** is operated as needed to move head section **80** to a raised position, motors **92, 94** are operated as needed to flatten or lower thigh and foot section **83, 84**, motors **100** are operated as need to move the upper frame **50** to its low position relative to base frame **48**, and if mattress **42** is an air mattress having its pneumatic system **98** controlled by control circuitry **110**, to operate the pneumatic system **98** to increase the pressure in a seat section of the mattress **42**. All of these motions and operations make it easier for the patient to stand up out of bed **12**. For example, by raising the head section **80**, the siderails **58** are moved into ergonomic positions for the patient grip while standing up.

(46) Field **186** includes a chair button **196** situated above button **192**. Button **196** is pressed to move deck **74**, and therefore mattress **42** supported by deck **74** and the patient supported by the mattress **42**, into a chair position. While button **196** is pressed, motor **90** is operated as needed to pivotably raise head section **80** relative to upper frame **50**, motor **92** is operated as needed to pivotably raise thigh section **83** relative to upper frame **50**, and motor **94** is operated as needed to pivotably lower foot section **94** relative to thigh section **83**. Field **186** also includes a 30 degree limit control button **198** for enabling and disabling a 30 degree limit function of bed **12**. When enabled, the 30 degree limit control prevents head section **80** of bed **12** from being lowered to a HOB angle less than 30 degrees. By keeping head section **80** raised above 30 degrees, some pulmonary complications such as ventilated assisted pneumonia (VAP) are inhibited from occurring. Sequential presses or touches of button **198** toggles the 30 degree limit function of bed **12** between enabled (i.e., on) and disabled (i.e., off) states.

(47) Control panel **106** of bed **12** includes a master lockout button that, when pressed, permits any of buttons **188, 190, 192, 194, 196, 198** to be simultaneously pressed to lockout the associated function of buttons **188, 190, 192, 194, 196, 198**. That is, when locked out, the respective button **188, 190, 192, 194, 196**, cannot be used as an input to cause the associated bed movements. Also, when locked out, button **198** cannot be used to enable the 30 degree lockout function. In essence, when button **198** is locked out, the 30 degree limit function is locked in its disabled state. A lockout icon **200** is displayed in field **186** adjacent to the respective buttons **188, 190, 192, 194, 196, 198** that have been locked out. However, a single lockout icon **200** is shown to indicate that buttons **188, 190** related to foot section **84** extension and retraction are both been locked out together. After any of buttons **188, 190, 192, 194, 196, 198** are locked out, they are unlocked in the same manner as just described. That is, the master lockout button on control panel **106** is pressed and then the locked out button **188, 190, 192, 194, 196, 198** to be unlocked is pressed simultaneously. The master lockout button on panel **106** is a “hard” or analog button such as a membrane switch, for example, whereas the buttons of home screen **144** are “soft” buttons that are implemented in software as is known in the art.

(48) Referring now to FIGS. **4A** and **4B**, a software algorithm **210** that is stored in memory **114** and executed by microprocessor **112** of control circuitry **110** of bed **12** is shown. Algorithm **210** includes the various steps, conditions, and user input selections that are implemented by bed **12** as a practical application for entering manual bed location data using GUI **120** of bed **12**. Algorithm **210** starts at block **212** and proceeds to block **214** to determine whether bed **12** is plugged in. Thus, at block **214** control circuitry **110** determines whether plug **166** is connected to AC outlet **170** such that current or voltage is sensed by a power circuitry module (not shown) of control circuitry **110**. The power circuitry module or circuitry **110** may include, for example, one or more transformers, rectifiers, voltage converters, voltage dividers, voltage regulators, resistors, inductors, capacitors, and the like, at the discretion of the bed designer.

(49) If it is determined at block **214** of FIG. **4A** that bed **12** is not plugged in, then algorithm **210** loops back to step **212** and proceeds from there. If it is determined at block **214** that bed **12** is plugged in, then algorithm **210** proceeds to block **216** where microprocessor **112** of control circuitry **110** determines whether casters **49** of bed **12** are braked or have their brakes locked against rolling based on signals from caster braking sensors **124**. If it is determined at block **216** that casters **49** are not locked, then algorithm **210** loops back to step **212** and proceeds from there.

(50) If it is determined at block **216** of FIG. **4A** that casters **49** of bed **12** are braked or locked, then algorithm **210** proceeds to block **218** at which microprocessor **112** of control circuitry **110** determines whether manual location has been completed by a caregiver using GUI **120** within 30 seconds. Thus, microprocessor **112** implements a software timer for 30 seconds to give the caregiver time to manually enter bed location data using GUI **120**. In this regard, when bed **12** is plugged into AC outlet **170** and casters **49** are braked, home screen **144** appears on GUI **120** but without window **142** appearing on screen **144** because there is not yet any room location associated with bed **12**, as shown in the upper right region of FIG. **4A**. In some embodiments, after bed **12** is plugged into outlet **170**, a reminder screen appears on GUI **120** to remind the caregiver to brake casters **49** using foot pedals **51**. In any event, once home screen **144** appears on GUI **120** as shown in FIG. **4A**, the caregiver is able to select location button **150** to navigate to subsequent screens for manual entry of bed location information or data as described below.

(51) If it is determined at block **218** that manual entry of bed location information has occurred within 30 seconds, then algorithm **210** ends as indicated at block **220**. In other embodiments, a time threshold greater than or less than 30 seconds may be implemented at block **218** of algorithm **210**. For example, a time threshold of 1 minute or even up to 2 minutes are within the scope of the present disclosure, just to give a couple of arbitrary examples. If it is determined at block **218** that location information has not been entered manually within 30 seconds of the bed **12** being plugged in and the casters **49** locked, then algorithm **210** proceeds to block **222** to play a voice prompt reminding the caregiver that bed location data needs to be manually entered using GUI **120**. The voice prompt is implemented, for example, as a sound file stored in memory **114** and played through a speaker (not shown) of bed **12**. The voice prompt includes a message such as, for example, “Bed not located. Please input location.” Other audio prompts are within the scope of the present disclosure at the discretion of the bed designer or programmer.

(52) After the voice prompt is played at block **222**, or substantially simultaneously with playing the voice prompt at block **222**, algorithm **210** causes a bed-not-located screen **224** to be displayed on GUI **120** as shown in FIG. **4A**. Screen **224** includes a close button or icon **226** that is selected if the caregiver does not wish to manually enter location data into memory **114** of bed **12** for communication via module **16** to network **18**. After button **226** is selected, home screen **144** is displayed on GUI **120**. Screen **224** also includes an OK icon or button **228** that is selected to navigate to further screens for manual entry of bed location data or information as will be discussed in further detail below. In the illustrative example, screen **224** also displays the text string, “BED NOT SYNCHED WITH NURSE CALL SYSTEM. INPUT BED LOCATION.”

(53) Giving caregivers a threshold period of time for entry of manual location data at block **218** of algorithm **210** is an improvement over the algorithm depicted in FIG. 11 of U.S. Patent Application Publication No. 2018/0039743 A1 in which manual entry screens are displayed in response to AC power being applied to the bed. For example, the caregiver may wish to see other information on home screen **144** regarding bed status prior to beginning the process of manually entering bed location information. Furthermore, requiring bed **12** to be plugged in to an AC outlet **170** and requiring casters **49** to be locked or braked in algorithm **210** prior to display of bed-not-located also represents an improvement over the algorithm depicted in FIG. 11 of U.S. Patent Application Publication No. 2018/0039743 A1 because the two conditions of algorithm **210** at blocks **214**, **216** provides redundant assurance that the bed **12** is intended to remain at its location for an extended period of time, thereby warranting manual entry of bed location information. Still further, the

algorithm of FIG. 11 of U.S. Patent Application Publication No. 2018/0039743 A1 does not include any voice prompt like the algorithm **210** of the present disclosure. The voice prompt at block **222** of algorithm **210** assures that the caregiver is alerted to the need for manual entry of bed location information if the caregiver does not otherwise notice the bed-not-located screen **224** being displayed on GUI **120**, for example.

(54) In response to OK icon **228** being selected on screen **224**, algorithm **210** causes a select-the-bed-location screen **230** being displayed on GUI **120** as shown in FIG. 4A. Screen **230** also appears on GUI **120** in response to location button **150** on home screen **144** being selected by the caregiver. Furthermore, screen **230** appears on GUI **120** in response to edit icon **148** of window **142**, shown in FIG. 3, being selected on home screen **144**. Thus, there are three ways in which screen **230** becomes displayed on GUI **120**, as follows: selection of OK button **228** on screen **224**; selection of location button **150** on screen **144**; and selection of edit icon **148** of window **142**, if shown, on screen **144**. Regardless of the manner in which the caregiver navigates to screen **230**, the remaining discussion below of screen **230** and the screens of FIG. 4B is equally applicable.

(55) Screen **230** includes a table **232** having a list of room locations that can be selected by touching the row in the table **232** corresponding to the room in which bed **12** is located. An up arrow icon **234** and a down arrow icon **236** are provided on screen **230** in the right hand portion of table **232**. Icons **234**, **236** are touched by the caregiver to scroll up or down, respectively, to view other room location choices that are available for selecting by the caregiver on table **232**. In the illustrative example of FIG. 4A, the location of MED/SURG **2011-A** is selected in table **232** as indicated by the border highlighting around the selected row in table **232**. A delete icon **238** is shown to the left of the selected bed location in table **232**. Beneath table **232** of screen **230** is a campus edit icon or button **240** and a unit edit button or icon **242**.

(56) At the bottom of screen **230** is a cancel button or icon **244** and a select icon or button **246**. If the caregiver does not wish to make any manual location selections, the cancel button **244** is selected and GUI **120** will return to showing home screen **144**. If the room location of table **232** is the desired room location and if the desired campus and unit selections appear to the right of respective buttons **240**, **242**, the select button **246** is selected to store the room location, campus location, and unit location information in memory **114** of control circuitry **110** of bed **12**. After the bed location information is stored in memory **114**, control circuitry **110** controls wireless communication module **16** of bed **12** to transmit the bed location information to one or more of servers **22**, **24**, **26** of the corresponding systems **23**, **25**, **27** of network **18** via one or more WAP's **14**.

(57) In some embodiments, module **16** of bed **12** transmits the stored location information a single time and, if an acknowledgement message is returned to module **16** of bed via one or more WAP's **14** indicating that the desired destination server(s) **22**, **24**, **26** has successfully received the location information from bed **12**, then module does not transmit the location information subsequently unless a change is made to the location information using GUI **120** or unless bed **12** is unplugged from outlet **170** for a threshold period of time in which case algorithm **210** starts anew at block **212**. In such embodiments, a bed ID such as a bed serial number or MAC address is transmitted with the bed location information. During subsequent bed status data transmissions from bed **12**, the bed ID is also transmitted and is used by the receiving server(s) **22**, **24**, **26** to correlate or associate the bed status information with the bed location information. In other embodiments, the bed location data is transmitted by module **16** of bed along with each transmission of bed status data after the bed location information has been stored in memory **114** of control circuitry **110**.

(58) If the caregiver wishes to select a different campus location while viewing screen **230**, the campus icon **240** is selected on screen **230** of FIG. 4A and algorithm **210** proceeds to display a select-the-campus-location screen **248** on GUI **120** as shown in FIG. 4B. The line labeled with circle A shows the screen flow of algorithm **210** in this regard. Screen **248** includes a table **250** having a list of campus locations that can be selected by touching the row in the table **250**

corresponding to the campus at which bed **12** is located. Up and down arrow icons **234**, **236** are provided at the right hand portion of table **250** and are touched to scroll up or down, respectively, to view other campus location choices that are available for selecting by the caregiver on table **250**. In the illustrative example of FIG. **4B**, the campus of ST. ELIZABETH SOUTH is selected in table **250** as indicated by the border highlighting around the selected row in table **250**.

(59) Delete icon **238** is shown to the left of the selected campus location in table **250**. Beneath table **250** of screen **248** is campus edit icon or button **240** and a unit edit button or icon **242**. At the bottom of screen **248** is cancel button or icon **244** and select icon or button **246**. Buttons **238**, **240**, **242**, **244**, **246** on screen **248** function in the same manner as these same buttons **238**, **240**, **242**, **244**, **246** function on screen **230**. For example, if the caregiver does not wish to make any manual location selections on screen **248**, the cancel button **244** is selected and GUI **120** will return to showing home screen **144** or, in some embodiments, screen **230**. If the campus location of table **250** is the desired campus location and if the desired unit selection also appears to the right of button **242**, the select button **246** is selected to store the campus location and unit location in memory **114** of control circuitry **110** of bed **12** for eventual transmission by module **16** as described above.

(60) If the caregiver wishes to select a different unit location while viewing screen **230** or while viewing screen **248**, the unit icon **242** is selected on screen **230** of FIG. **4A** or screen **248** of FIG. **4B**, as the case may be, and algorithm **210** proceeds to display a select-the-unit-location screen **252** on GUI **120** as shown in FIG. **4B**. The line labeled with circle B shows the screen flow of algorithm **210** in this regard in connection with selection of button **242** on screen **230**. Screen **252** also appears on GUI **120** in response to the caregiver selecting button **246** on screen **248**. In either case, screen **252** includes a table **254** having a list of unit locations that can be selected by touching the row in the table **254** corresponding to the unit at which bed **12** is located. Up and down arrow icons **234**, **236** are provided at the right hand portion of table **254** and are touched to scroll up or down, respectively, to view other unit location choices that are available for selecting by the caregiver on table **254**. In the illustrative example of FIG. **4B**, the unit of WEST TOWER is selected in table **254** as indicated by the border highlighting around the selected row in table **254**.

(61) Delete icon **238** is shown to the left of the selected unit location in table **254**. Beneath table **254** of screen **252** is campus edit icon or button **240** and unit edit button or icon **242**. At the bottom of screen **252** is cancel button or icon **244** and select icon or button **246**. Buttons **238**, **240**, **242**, **244**, **246** on screen **252** function in the same manner as these same buttons **238**, **240**, **242**, **244**, **246** function on screens **230**, **248**. For example, if the caregiver does not wish to make any manual location selections on screen **252**, the cancel button **244** is selected and GUI **120** will return to showing home screen **144** or, in some embodiments, return to showing screen **230**. If the unit location of table **254** is the desired unit location and if the desired campus selection also appears to the right of button **240**, the select button **246** of screen **252** is selected to store the campus location and unit location in memory **114** of control circuitry **110** of bed **12** for eventual transmission by module **16** as described above.

(62) As noted above with regard to screen **230**, if the room location of table **232** is the desired room location and if the desired campus and unit selections appear to the right of respective buttons **240**, **242**, the select button **246** is selected to store the room location, campus location, and unit location information in memory **114** of control circuitry **110** of bed **12**. Selection of button **246** on screen **230** of FIG. **4A** results in a modified start-up screen **140'** appearing on GUI **120** as shown in FIG. **4B**. The line labeled with circle C shows the screen flow of algorithm **210** in this regard in connection with selection of button **246** on screen **230**. On screen **140'**, window **142** now shows text string **146** with the selected room location information, MED/SURG **2011-A** in the illustrative example, that was chosen using table **232** of screen **230**. Window **142** appears on screen **140'** for a threshold period of time, such as 5 to 30 seconds, and then disappears as was described above.

(63) While window **142** appears on screen **140'**, the caregiver has the option of selecting edit icon

148 to further edit the bed location information if desired. In response to selection of button **148** on screen **140'**, algorithm **210** causes a current-bed-location screen **256** to appear on GUI **120** as shown in FIG. **4B**. Screen **256** includes a table **258** that indicates with border highlighting the current bed location that was selected previously. The caregiver can then make another selection on table **258** to change the room location information. Scroll arrows **234**, **236** are provided in table **258** for use by the caregiver to scroll other room location options as noted above. Buttons **240**, **242**, **244**, **246** appear on screen **256** and function in the same manner as described above. For example, selection of button **240** results in navigation to screen **248**, selection of button **242** results in navigation to screen **252**, selection of button **244** results in navigation back to screen **140'**, and selection of button **246** results in navigation back to screen **140'** but with the new room location information manually selected on table **258** appearing as text string **146** in window **142**.

(64) In response to button **238** being selected on screen **256** of FIG. **4B**, algorithm **210** causes screen **230** to, once again, be shown on GUI **120**. The description above of screen **230** shown in FIG. **4A** is equally applicable to screen **230** shown in FIG. **4B**. Selection of button **238** on screen **230**, be it the one shown in FIG. **4A** or the one shown in FIG. **4B**, results in a location delete screen **260** appearing on GUI **120**. Thus, button **238** is selected when the caregiver wishes to delete a room location from even appearing on table **232**. Screen **260** includes the following textual information: "ARE YOU SURE YOU WANT TO DELETE THIS LOCATION? RESTORING A LOCATION MUST BE DONE IN NURSE CALL SYSTEM." Screen **260** includes a yes button **262** that is selected if the caregiver does, indeed, wish to delete the selected bed location information from table **232** altogether. Screen **260** also includes a no button **264** that is selected to abort the deletion of the selected of the bed location information selected on table **232**.

(65) After either of buttons **262**, **264** is selected on screen **260**, algorithm **210** returns back to screen **230** so that the caregiver can, if desired, select the bed location information from among the remaining information on table **232** and then select icon **246**, or the user can select icon **244** and return to screen **140'** or screen **144** as the case may be. Button **238** of screen **248** and button **238** of screen **252** operate in a substantially similar manner as button **238** of screen **230** except that selected campus location information or selected unit location information, as the case may be, are deleted from the respective tables **250**, **254** by navigating to screen **260** and selecting the yes button **262**.

(66) Referring now to FIGS. **5A-5C**, a parameters screen **300** is shown on GUI **120**. GUI **120** is embodied as a touch screen display having various icons or buttons that are selectable by a caregiver to navigate to other screens for selection of various bed functions and entry of various types of data as will be discussed below, primarily in connection with manual entry of patient identification data on GUI **120**. Parameters screen **300** appears on GUI **120**, in some embodiments, in response to a caregiver (or really, any user for that matter) touching GUI **120** to wake GUI **120** from a dormant state in which GUI **120** is blacked out or, in some embodiments, in which GUI **120** displays a screen saver image of some sort. In some embodiments, the parameters screen **300** is displayed by selecting a parameters button on a home screen or other screen. In some embodiments, the control circuitry **110** is configured to play a voice prompt to remind the caregiver to manually validate the patient identification data after a threshold period of time elapses after a remote server, such as an ADT server, an EMR server, bed status data server, or the like has communicated the patient name to the bed **12**. In some embodiments, the threshold period of time begins after the patient enters the bed **12**. In some embodiments, the threshold period of time begins after the caregiver awakens the GUI **120** from a sleep screen.

(67) The parameters screen **300** includes a field **302** having icons and buttons that when selected navigate the user to various screens. A home button **304** is selectable, such as by touching, to navigate to a home or start screen. A weigh scale button **306** is selectable, such as by touching, to navigate to the screens that are used to provide information on a weight of the patient as measured by the scale system **70**. An alarm button **308** is selectable, such as by touching, to navigate to an

alarm screen that enables the user to set various alarms for the bed **12**, e.g. a bed exit alarm. A settings button **310** is selectable, such as by touching, to navigate to various screens that enable bed settings to be adjusted, e.g. bed position settings.

(68) The parameters screen **300** also includes a parameters field **320** having icons and buttons that, when selected, allow the user to adjust various parameters of the bed. For example, a settings button **322** is selectable, such as by touching, to navigate to various screens that enable the user to adjust various settings related to patient identification. A software verification button **324** is selectable, such as by touching, to navigate to a software screen where a user can verify that the software operating the control circuitry **110** is updated, loaded, and/or operational. A new patient button **326** is selectable, such as by touching, to navigate to a screen that allows the user to enter patient information related to a newly admitted patient, e.g. name, address, patient ID number, etc. A maintenance button **328** is selectable, such as by touching, to navigate to various maintenance screens that enable a technician to trouble-shoot and correct technical failures of the bed **12**. A failure codes button **340** is selectable, such as by touching, to display a list of failure codes related to the bed **12**. Such failure codes may be utilized to identify errors in the bed **12**. A technician access button **342** is selectable, such as by touching, to provide a technician access to the software of the bed **12** so that maintenance can be performed. The technician access button **342** also allows a technician to set specific parameters related to the various functions of the bed **12**, as described herein. A WiFi button **344** is selectable, such as by touching to navigate to a WiFi screen **360**, wherein WiFi connectivity is validated.

(69) Referring still to FIG. 5A, the WiFi screen **360** includes a bed field **362** that displays an identification number associated with the bed **12**. Accordingly, a user may read this field to verify that the correct bed **12** is in use. A connectivity field **364** includes various icons that confirm connectivity of the bed **12**. A power button **366** is selectable to turn the WiFi functions of the bed **12** on and off. The power button **366** is illuminated in a first color, e.g. green, to verify that the WiFi is turned on. The power icon **366** may be unilluminated or illuminated in a second color, e.g. red, if the WiFi is not turned on. A WiFi connection icon **368** reads “on” when the wireless communication module **16** is powered. The WiFi connection icon **368** may read “off” when the wireless communication module **16** is not powered. A signal strength icon **370** includes a plurality of bars that are illuminated to indicate a strength of a WiFi network connection. For example, the number of bars illuminated is directly proportional to a strength of the signal. That is, the more bars that are illuminated, the stronger the signal is. A server icon **372** is illuminated in a first color, e.g. green, to indicate that the bed **12** is connected to a server. If the bed **12** is not connected to a server, the server icon **372** may be unilluminated or illuminated in a second color, e.g. red.

(70) A location icon **374** is illuminated in a first color, e.g. green, if bed location has been communicated to the bed **12** by the real-time locating system **25** or by manual entry of bed location using GUI **120** of the bed **12**. The location icon **374** may be unilluminated or illuminated in a second color, e.g. red, if the location of the bed **12** has not been communicated to the bed **12** by the real-time locating system **25** or the bed **12** has not been located by manual entry. A patient identification icon **376** is provided to notify the caregiver whether the patient on the bed **12** is identified or validated. If the patient has been identified and/or validated, the patient identification icon **376** is illuminated in a first color, e.g. green. If the patient has not been identified or validated, the patient identification icon **376** is unilluminated or illuminated in a second color, e.g. red. The patient being validated involves a manual validation process by a user as will be described below.

(71) A patient identification field **390** lists a name or patient number or other patient identifier of the patient on the bed **12**. If the patient has not been identified, the patient identification field **390** is blank or simply lists the room number in which bed **12** is located as shown in FIG. 5A. WiFi screen **360** includes a data field **400** that includes various icons and buttons for selecting the location of the bed **12** and the identification of the patient. For example, a locate button **402** is selectable, such as by touching, to manually identify a room in which the bed is positioned, as described above. A

patient ID button **404** is selectable, such as by touching, to navigate to a patient identification screen **420**. A cancel button **406** is selectable, such as by touching, to return to the parameters screen **300**. The patient identification screen **420** includes text **422** asking the user to verify or confirm that the listed patient is the actual patient on the bed **12**. For example, the text **422** read “DO YOU CONFIRM THAT THIS PATIENT IS ON THE BED?” in the illustrative example. Below the text **422** the name of a patient and/or a patient ID number is listed.

(72) The listed name of the patient below text **422** may be the name of the patient most recently received from a hospital server. In some embodiments, the listed name of the patient may be the name of the last patient to have been confirmed to be on the bed **12**. In other embodiments, the listed patient name is the name of a patient that has been newly assigned to the bed **12** and communicated to the bed **12** from a remote server such as an ADT server, EMR server **26**, bed status data server, or the like. In some embodiments, the patient room assignment for each patient is entered at an ADT computer at the time of patient admission and then, the associated ADT server sends the patient identification (ID) and location ID to the bed status data server which, in turn, determines the bed ID of the bed **12** at the location corresponding to the location ID. The bed status data server then communicates the patient ID to the bed **12** for inclusion on the screen **420** beneath the text **422**.

(73) A yes button **430** is selectable, such as by touching, to verify/confirm that the patient listed in the text **422** is the patient on the bed **12**. Selecting the yes button **430** returns the user to the WiFi Screen **360**, as shown in FIG. 5B. As illustrated in FIG. 5B, at this stage, the patient name and identification number is shown in the patient identification field **390**. Also, the patient identification icon **376** is illuminated in the first color. FIG. 5B also illustrates a home screen **440** that is displayed after the home button **304** is selected. Once the patient identification is verified, the home screen **440** illuminates a patient identification icon **442**, similar to patient identification icon **376**.

(74) The patient identification screen **420** also includes a cancel button **450** that is selectable, such as by touching, when the identification of the patient cannot be confirmed or, for some reason, the caregiver or user decides not to confirm the identification of the patient. In some embodiments, the cancel button **450** may be a no button. Selecting the cancel button **450** returns the user to the WiFi Screen **360**, as shown in FIG. 5C. As illustrated in FIG. 5C, at this stage, the patient name and/or identification number are not shown in the patient identification field **390** but the room number still appears in the field **390**. Also, the patient identification icon **376** is either unilluminated or illuminated in the second color. FIG. 5C also illustrates the home screen **440** that is displayed after the home button **304** is selected. When the patient identification is not verified, the home screen **440** does not illuminate the patient identification icon **442** or the patient identification icon **442** is illuminated in the second color.

(75) FIG. 6 illustrates a weight input screen **500** that includes a current weight field **502**. The current weight field **502** displays a current weight of the patient on the bed **12** as measured by the scale system **70**. A saved weight field **504** displays a previously saved weight of the patient on the bed **12**. The current weight and the saved weight are displayed in kilograms; however, the settings of the GUI **120** may be altered to display the current weight and the saved weight in other units such as pounds. A variance field **506** displays a change in the patient's weight from the saved weight to the current weight. Accordingly, the caregiver may track and record changes in the patient's weight.

(76) An input field **520** includes various icons and buttons that are selectable, such as by touching, to record data related to the patient's weight. A close button **522** closes the weight input screen **500** and returns the user to the home screen **440** or a previously opened screen. A height button **524** is selectable to enter the height of the patient. Accordingly, the height of the patient may be saved in electronic medical record **27**. Additionally, by entering the patient height and patient weight, circuitry **110** of bed **12** is able to calculate a body mass index (BMI) of the patient on the bed **12**.

(77) A save current weight button **530** is selectable, to save the current weight in the electronic

medical record system **27**. By selecting the save weight button **530**, one of two screens is displayed depending upon whether the identification of the patient has been verified prior to taking the patient's weight. If the patient has been verified, as set forth above, the save weight screen **540** is displayed on the GUI **120**. The save weight screen **540** includes a patient identification icon **542** that is illuminated in a first color, e.g. green, to indicate that the weight is being saved for the verified patient. A text field **544** displays text that reads "VERIFY THAT ONLY ITEMS PRESENT DURING ZEROING ARE ON BED." The text field also displays the patient name and, in some embodiments, patient identification number. Accordingly, the caregiver may select a save button **546** to take a verified patient weight reading when the patient identification has been verified. Conversely, the caregiver may select a cancel button **548** if other objects have been added to, or removed from, the bed **12** subsequent to zeroing or taring the scale system **27**, or if the patient listed is not the correct patient.

(78) Selection of the save button **546** results in transmission of the weight data tagged with the patient identification data to the electronic medical record system **27** for storage in the patient's electronic medical record. Selection of the save button **546** also results in storing, locally, in the bed memory, the patient weight thereby allowing the current weight to be compared to the previously stored weight. In some embodiments, the weight and verified patient ID data is transmitted first to the bed status data server and then from the bed status data server to the EMR server **26** of system **27**. In some embodiments, module **16** of bed **12** transmits the weight data tagged with the patient identification data a single time and, if an acknowledgement message is returned to module **16** of bed via one or more WAP's **14** indicating that the electronic medical record system **27** has successfully received the weight data tagged with the patient identification data from bed **12**, then module **16** does not transmit the location information subsequently unless a change is made to the weight data or to the patient identification data using GUI **120** or unless bed **12** is unplugged from outlet **170** for a threshold period of time in which case the patient identification data and the weight data must be reverified. In such embodiments, a bed ID such as a bed serial number or MAC address is transmitted with the weight data tagged with the patient identification data. During subsequent bed status data transmissions from bed **12**, the bed ID is also transmitted and is used by the bed status data server and/or the electronic medical server **26** to correlate or associate the weight data tagged with the patient identification data with the bed location information. In other embodiments, the weight data tagged with the patient identification data is transmitted by module **16** of bed **12** along with each transmission of bed status data after the bed location information has been stored in memory **114** of control circuitry **110**.

(79) If the patient has not been identified, as set forth above, the save weight screen **560** is displayed on the GUI **120** in response to selection of the button **530** on the screen **500**. The save weight screen **560** includes the patient identification icon **542** that is unilluminated or illuminated in a second color, e.g. red, to indicate that the weight is being saved for an unidentified patient. A text field **544** displays text that reads "VERIFY THAT ONLY ITEMS PRESENT DURING ZEROING ARE ON BED." In this scenario, the text field **544** does not identify a patient. Accordingly, the caregiver may select a save button **566** to take an unverified weight reading when the patient identification has not been verified. Conversely, the caregiver may select a cancel button **568** if other objects have been added to, or removed from, the bed **12** subsequent to zeroing or taring the scale system **27**.

(80) Selection of the save button **566** results in transmission of the weight data without patient identification data to the bed status data server and/or to EMR server **27** of the EMR system **27**. In some embodiments, the weight data may be tagged with location identification data or bed identification data. In some embodiments, module **16** of bed **12** transmits the weight data a single time and, if an acknowledgement message is returned to module **16** of bed via one or more WAP's **14** indicating that the electronic medical record system **27** has successfully received the weight data from bed **12**, then module does not transmit the location information subsequently unless a change

is made to the weight data using GUI **120** or unless bed **12** is unplugged from outlet **170** for a threshold period of time in which case the weight data must be reverified. In such embodiments, a bed ID such as a bed serial number or MAC address is transmitted with the weight data. During subsequent bed status data transmissions from bed **12**, the bed ID is also transmitted and is used by the bed status data server and/or the EMR server **26** to correlate or associate the weight data with the bed location information. In other embodiments, the weight data is transmitted by module **16** of bed **12** along with each transmission of bed status data after the bed location information has been stored in memory **114** of control circuitry **110**.

(81) Referring now to FIG. **7**, another example of a WiFi screen **600** includes a text box **602** that displays the bed identification number. A connectivity field **610** includes various icons that confirm connectivity of the bed **12**. A power button **612** is illuminated in a first color, e.g. green, to verify that the bed **12** is connected to a WiFi network. The power button **612** may be unilluminated or illuminated in a second color, e.g. red, if the bed **12** is not connected to a WiFi network. WiFi connection icon **614** reads “on” when the wireless communication module **16** is powered. The WiFi connection icon **614** may read “off” when the wireless communication module **16** is not powered. A signal strength icon **616** includes a plurality of bars that are illuminated to indicate a strength of the network connection. For example, the number of bars illuminated is directly proportional to a strength of the signal. That is, the more bars that are illuminated, the stronger the signal is. A server icon **618** is illuminated in a first color, e.g. green, to indicate that the bed **12** is connected to a server. If the bed **12** is not connected to server the server icon **618** may be unilluminated or illuminated in a second color, e.g. red. A location icon **620** is illuminated in a first color, e.g. green, if bed location has been communicated to the bed **12** by the real-time locating system **25** or by manual entry of bed location. The location icon **620** may be unilluminated or illuminated in a second color, e.g. red, if the location of the bed **12** has not been communicated to the bed **12** by the real-time locating system **25** or the bed **12** has not been located by manual entry.

(82) A room text box **630** displays the room number in which the bed **12** is positioned. If a room number has not been assigned to the bed **12**, the room text box **630** is blank. A cancel button **632** is selectable, such as by touching, to cancel the WiFi screen **600** and return to the home screen **440**. A bed location button **634** is selectable, such as by touching to initiate the process of associating a room number to the bed **12**. Selection of the bed location button **634** results in a room selection screen **640** having a list **642** of rooms in the healthcare facility being displayed on the GUI **120**. By selecting an available button **644**, all rooms available in the healthcare facility are displayed in the list **642**. The list may be scrolled through by scrolling with a finger on the GUI **120**. If desired, up and down arrow buttons provided on the screen **640** to the right of the list **642** may be used for scrolling instead. Selecting an assigned button **646** populates the list **642** with all rooms in the healthcare facility that have been assigned to respective patients. The user may select a room number from the list **642**, such as by touching the screen. An enlarge button **650** is selectable, such as by touching, to navigate one level up in a hierarchy, e.g. room, floor, building. A cancel button **652** is selectable, such as by touching, to return to the WiFi screen **600**, the home screen **440**, or any other previously displayed screen.

(83) Selection of a room number from the list **642** navigates the user to a patient identification screen **660**. The patient identification screen **660** includes a text box **662** that reads, “DO YOU CONFIRM THAT THIS PATIENT IS ON THE BED?” A patient name is provided beneath the text box **662**. The user can select, such as by touching, a cancel button **664** if the user cannot confirm the identification of the patient on the bed **12** or, for some reason, the caregiver or user decides not to confirm the identification of the patient. Selection of the cancel button **664** returns the user to a previous screen without patient identification data. In some embodiments, the cancel button **664** is a no button. Selection of a yes button **668**, for example by touching, confirms that the named patient is on the bed **12** and results in the transmission of patient identification data to the electronic medical record system **27** either directly or via the bed status data server.

(84) Referring now to FIG. 8, a change configuration screen **700** is displayed on GUI **120** in response to selecting the settings button **322** on the parameters screen **300**. The change configuration screen **700** includes a field **702** that reads, "WIFI MODULE CONNECTED." A yes button **704** and a no button **706** are displayed in the field **702**. Selection of the yes button **704** results in the wireless module **16** connecting to the communication infrastructure **18** in which case fields **368**, **614** indicate "WIFI ON." Selection of the no button **706** disconnects the wireless module **16** from the communication infrastructure **18** in which case fields **368**, **614** indicate "WIFI OFF." Selection of the no button **706** also disables all screen related to connectivity. Selection of the no button **706** means that the bed is not equipped with a communication module, therefore, all connectivity screens would have no effect. A module (when present) can be turned on or off using button **366** (shown in FIG. 5A). The change configuration screen **700** includes another field **710** that reads, "MANUAL LOCATING." An on button **712** and an off button **714** are displayed in the field **710**. Selection of the on button **712** configures the bed **12** to permit the caregiver to select the bed location manually as described above. Selection of the off button **714** disables the manual location selection feature from being usable on the bed **12**. The change configuration screen **700** also includes a field **720** that reads, "PATIENT ID DISPLAY." A yes button **722** and a no button **724** are displayed in the field **720**. Selection of the yes button **722** causes the GUI **120** to display the patient identification icon **376** for use in verifying the patient identification as described above. Selection of the no button **724** causes the patient identification icon **376** to not be displayed on the GUI **120** thereby disabling the patient identification verification feature from being usable on the bed **12**. A Save button **730** is selectable, such as by touching, to save the settings.

(85) In some embodiments, the user may enter any one of location data, patient identification data, and weight data using the GUI **120** of the bed **12**. For example, the user may enter one of location data, patient identification data, and weight data, in some embodiments. In some embodiments, the user may enter any two of location data, patient identification data, and weight data. In some embodiments, the user may enter all three of location data, patient identification data, and weight data.

(86) In some embodiments, the patient identification data may be temporarily invalidated. For example, the patient identification data may be temporarily invalidated in response to the wireless connection being lost, e.g. if the wireless connection is lost for over a predetermined period of time. In some embodiments, the patient identification data may be temporarily invalidated in response to the bed **12** being unplugged from the outlet **170** of the healthcare facility, e.g. if the bed **12** is unplugged from the outlet for longer than a predetermined period of time such as about ten seconds to about one minute. Optionally, the patient identification data may be temporarily invalidated in response to the patient exiting the bed **12**, e.g. if the patient has exited the bed **12** for longer than a predetermined period of time.

(87) When the patient identification data is temporarily invalidated, the patient identification data may be revalidated upon the conclusion of the temporary invalidating event. For example, if the wireless connection is reconnected, the patient identification data may be revalidated. In another example, if the bed is plugged back in, the patient identification data may be revalidated. In some embodiments, if the patient reenters the bed **12**, the patient identification data may be revalidated. In some embodiments, the caregiver is prompted to revalidate the patient identification data. For example, the caregiver may be directed to a location revalidation screen **800**, as illustrated in FIG. 9. The revalidation screen **800** includes a text box **802** that states the room that the bed **12** was previously plugged into and asks the caregiver to confirm whether the bed **12** is still plugged into that room. A confirm button **804** is selectable to confirm that the bed is still plugged into the stated room. A different room button **806** navigates the user to the screen **640** (shown in FIG. 7) to enable the user to select another room. A cancel button **808** returns the user to a home screen. In some embodiments, the revalidation screen is similar to the WiFi screen **360**. In some embodiments, an alarm is triggered if the bed **12** is disassociated from its location.

(88) In some embodiments, the patient identification data may be permanently invalidated. That is, the patient identification data may be invalidated in response to an invalid patient ID being entered. The patient identification data of a previously validated patient may be invalidated during the process of entering a new patient identification using the new patient button **326**, in some embodiments. In some embodiments, the patient identification data may be invalidated in response to unplugging the bed **12** from AC power such as when the bed **12** is being transported, e.g. moved to another room. Optionally, the patient identification data of a previously validated patient may be invalidated when a new patient ID becomes available such as if a new patient is assigned to the bed **12** using an ADT system computer and the bed **12** receives the new patient data. If desired, the patient identification data is invalidated in response to new localization data being available from the real-time locating system **25** or entered manually on the GUI **120** of bed **12**. In some embodiments, the patient identification data may be invalidated in response to a configuration of the settings being changed using the change configuration screen **700**. The patient identification data may be invalidated in response to the patient exiting the bed **12** for longer than a predetermined period of time such as, for example, about 2 minutes to about 20 minutes such as for toileting, in some embodiments. After invalidation, the caregiver is prompted to revalidate the patient identification data. For example, the caregiver may be directed to the location revalidation screen **800**. In some embodiments, the revalidation screen is similar to the WiFi screen **360**. In some embodiments, an alarm is triggered if the bed **12** is disassociated from its location.

(89) Although certain illustrative embodiments have been described in detail above, variations and modifications exist within the scope and spirit of this disclosure as described and as defined in the following claims.

Claims

1. A patient bed comprising a frame configured to support a patient, circuitry carried by the frame, and a graphical user interface (GUI) carried by the frame and coupled to the circuitry, the circuitry being configured to command the graphical user interface to display at least one user interface screen that is usable by a caregiver to manually enter location data indicative of a location in a healthcare facility at which the patient bed is located, the circuitry being configured to transmit the location data entered by the caregiver and a bed identification (ID) from the bed, wherein the circuitry is configured to command the GUI to display a first screen of the at least one interface screen that is used by the caregiver to manually enter location data, wherein the circuitry is configured to prompt the caregiver to manually enter the location data of the patient bed after a threshold period of time elapses subsequent to a power plug of the circuitry being plugged into an outlet of a healthcare facility and subsequent to casters of the frame being braked.
2. The patient bed of claim 1, wherein the location data comprises a room number of the healthcare facility.
3. The patient bed of claim 1, wherein the GUI includes a change location button that is selectable by the caregiver to initiate a change of the location data prior to the threshold period of time elapsing.
4. The patient bed of claim 3, wherein a room menu screen appears on the GUI in response to the change location button being selected, the room menu screen permitting the caregiver to select a room number from a list of room numbers.
5. The patient bed of claim 3, wherein a campus icon and a unit icon appears on the GUI in response to the change location button being selected, the campus icon being selectable to cause a campus menu list being displayed on the GUI and the unit icon being selectable to cause a unit menu list being displayed on the GUI.
6. The patient bed of claim 1, wherein the first screen appearing on the GUI after the threshold period of time elapses includes a button that is selectable to initiate manual entry of the location

data.

7. The patient bed of claim 6, wherein a room menu screen appears on the GUI in response to the button being selected on the first screen, the room menu screen permitting the caregiver to select a room number from a list of room numbers.
8. The patient bed of claim 6, wherein a campus icon and a unit icon appears on the GUI in response to the button being selected on the first screen, the campus icon being selectable to cause a campus menu list being displayed on the GUI and the unit icon being selectable to cause a unit menu list being displayed on the GUI.
9. The patient bed of claim 1, wherein the circuitry includes a wireless communication module configured to wirelessly transmit the location data and the bed ID to a wireless access point for delivery to at least one remote computer for purposes of making a bed-to-room association.
10. The patient bed of claim 9, wherein the wireless communication module also is configured to transmit bed status data from the bed.
11. The patient bed of claim 1, wherein the circuitry is configured to play a voice prompt to remind the caregiver to manually enter the location data after the threshold period of time elapses.
12. The patient bed of claim 1, wherein the circuitry is configured to command the GUI to display a patient verification screen that is usable by the caregiver to verify an identity of a patient supported on the patient bed.
13. A patient bed comprising a frame configured to support a patient, circuitry carried by the frame, and a graphical user interface (GUI) carried by the frame and coupled to the circuitry, the circuitry being configured to command the graphical user interface to display at least one user interface screen that is usable by a caregiver to manually enter location data indicative of a location in a healthcare facility at which the patient bed is located, the circuitry being configured to transmit the location data entered by the caregiver and a bed identification (ID) from the bed, wherein the circuitry is configured to annunciate a voice prompt to the caregiver to manually enter the location data of the patient bed after a threshold period of time elapses subsequent to a power plug of the circuitry being plugged into an outlet of a healthcare facility and subsequent to casters of the frame being braked.
14. The patient bed of claim 13, wherein the circuitry also commands the GUI to display a first screen of at least one interface screen that is used by the caregiver to manually enter location data after the threshold period of time elapses.
15. The patient bed of claim 14, wherein the location data comprises a room number.
16. The patient bed of claim 14, wherein the first screen appearing on the GUI includes a button that is selectable to initiate manual entry of the location data.
17. The patient bed of claim 16, wherein a room menu screen appears on the GUI in response to the button being selected on the first screen, the room menu screen permitting the caregiver to select a room number from a list of room numbers.
18. The patient bed of claim 16, wherein a campus icon and a unit icon appears on the GUI in response to the button being selected on the first screen, the campus icon being selectable to cause a campus menu list being displayed on the GUI and the unit icon being selectable to cause a unit menu list being displayed on the GUI.
19. The patient bed of claim 15, wherein the circuitry includes a wireless communication module configured to wirelessly transmit the location data and the bed ID to a wireless access point for delivery to at least one remote computer for purposes of making a bed-to-room association.
20. The patient bed of claim 19, wherein the wireless communication module also transmits bed status data from the bed.
21. The patient bed of claim 14, wherein the circuitry is configured to command the GUI to display a patient verification screen that is usable by the caregiver to verify an identity of a patient supported on the patient bed.
22. The patient bed of claim 13, wherein the GUI includes a change location button that is

selectable by the caregiver to initiate a change of the location data prior to the threshold period of time elapsing.

23. The patient bed of claim 22, wherein a room menu screen appears on the GUI in response to the change location button being selected, the room menu screen permitting the caregiver to select a room number from a list of room numbers.

24. The patient bed of claim 22, wherein a campus icon and a unit icon appears on the GUI in response to the change location button being selected, the campus icon being selectable to cause a campus menu list being displayed on the GUI and the unit icon being selectable to cause a unit menu list being displayed on the GUI.
